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Designing energy sharing in energy communities: Review of coordination mechanisms and internal tariff structures

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ABSTRACT

Energy communities, also referred to as collective self-consumption, have rapidly emerged as a key organizational model for integrating renewable energy systems. While the literature has extensively addressed regulatory frameworks, business models and social dimensions, less attention has been given to the analytical foundations of energy sharing mechanisms and their optimization properties. This review provides a structured and unified examination of energy sharing within energy communities, focusing on allocation rules, optimization-based coordination, distributed algorithms, market-based paradigms, and internal tariff design. Clarifying this conceptual alignment enables a systematic comparison of coordination paradigms in terms of allocative efficiency, incentive compatibility, scalability, fairness integration, and regulatory embedding. The analysis further disentangles technical performance indicators from welfare-based efficiency and individual economic gains, revealing structural trade-offs that are often conflated in existing studies. Particular attention is devoted to internal tariff design and cost allocation principles, highlighting how pricing rules shape both economic outcomes and distributive properties. By integrating optimization theory, equilibrium modelling, and regulatory constraints, this review establishes a coherent framework for assessing and designing energy sharing architectures, while highlighting key research gaps in mechanism design, fairness integration, and implementation feasibility.

Nomenclature

Abbreviations	
ADMM	Alternating Direction Method of Multipliers
CSC	Collective Self-Consumption
DG	Distributed Generation
DSO	Distribution System Operator
EC	Energy Community
EMS	Energy Management System
ESh	Energy Sharing
ESS	Energy Storage System
EV	Electric Vehicle
GNE	Generalized Nash Equilibrium
ISC	Individual Self-Consumption
LP, QP	Linear/Quadratic programming
MILP	Mixed Integer Linear Programming
P2P	Peer-to-Peer
PV	Photovoltaic
RB	Rule-Based
RES	Renewable Energy Systems
SC	Self-Consumption
SCL	Solar Coverage Level
SCR	Self-Consumption Rate
SE	Shared Energy
SG	Smart Grid
SMG	Smart Micro Grid
SSR	Self-Sufficiency Rate
V2G	Vehicle-to-Grid
VI	Variational Inequality
VNB	Virtual Net Billing
Symbols	
$argmin_x f(x)$	Value of x when f reaches its minimum
C_i, C_{grid}	Cost function associated with user i/the grid
$E_{ij}(t)$	Exchanged energy between users i and j at time t
$E^{ch}(t), E^{dis}(t)$	ESS charge/discharge at time t
$E^{community,net}(t)$	Net exchanged energy between EC and grid
$E_i^{gen}(t)$	Energy generation from the user i at time t
$E_{tot}^{gen}(t)$	Total EC energy generation at time t
$E_i^{grid}(t)$	Net exchanged energy between user i and the grid at time t

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(*continued*)

$E_i^{grid,+}(t), E_i^{grid,-}(t)$	Energy imported from/exported to the grid by participant i at time t
$E_i^{load}(t)$	Energy demand from the user i at time t
$E_i^{share}(t)$	Shared energy allocated to the user i at time t
N, T	Participants and time sets
p, p^*	Price vector, clearing price
P^{reg}	Admissible pricing set
$SOC(t)$	State of Charge (in energy) of ESS at time t (in kWh)
SOC^{min}, SOC^{max}	Capacity bounds of ESS (in KWh)
U_i	Utility function associated with user i
w_{ij}	Matching coefficient between users i and j
x_i, X_i	Decision variable/feasibility sets associated with user i
x^{corr}, Π_F	Value of x^{eq} corrected by the projection Π_F onto a feasible set
x^{dual}	Dual optimization variable, associated to a dual problem
x^{eq}	Value of x when GNE is reached
x^{RB}, x^*	Rule-based allocation, centralized optimum
$\alpha_i(t)$	Allocation coefficient value at time t
η_c, η_d	ESS charge/discharge efficiency
ϕ_i	Shapley value for user i
$\lambda(t), \lambda^*(t)$	Lagrangian multiplier, optimal multiplier
$\pi^{buy}(t), \pi^{sell}(t), \pi^{comm}(t)$	Retail import, export remuneration and internal community transfer tariffs at time t
$\mathscr{L}$	Lagrangian

1. Introduction and general overview of energy communities

The increasing share of intermittent RES in this production has forced researchers, engineers, and policymakers to find solutions that make their management easier and thus avoid hindering their development [1]. It is recognized that facilitating the management of these intermittent and random productions will require: the development of energy storage systems (ESS) to shift and coincide production and consumption, that of smart grids (SG) to enable more efficient management of energy flows, and the development of load and production forecasting algorithms to anticipate events.

There is a revolution in the development of electricity production and distribution networks. The electrical grids were built to distribute electricity from large production centres to consumers with a passive role i.e., only receiving and consuming energy. With the significant growth of renewable production, both intermittent and uncontrollable, major challenges to electricity system operation have emerged [2]. Today and in the future, electricity networks will need to accommodate bidirectional power flows, across the territory while maintaining real-time supply-demand balance (which is particularly challenging given the high variability of RES generation). It is a new paradigm with distributed generation (DG) [3] which needs flexibility, continuous adaptability and dynamic processing of large volumes of data to determine the state of the system in real-time, made possible today thanks to the development of new information technologies [4]. In future networks, more electricity will be exchanged over short distance or stored for later use and less electricity will flow over long distances through large transmission networks. Since lower electricity volumes are transported, network cost will decrease which may reduce the retail electricity price [5].

Historically, the network only experienced downward flows, from centralized production connected at high voltage to low-voltage networks to serve end consumers. While these flows still largely remain the majority, the development of decentralized RES is generating upward flows, which will intensify significantly with a massive development of photovoltaic (PV) installations operating for self-consumption (SC) [6] (Fig. 1).

Increasing the share of intermittent production requires the development of forecasting techniques, grid-forming inverters, ESS and real-time EMS to maintain system stability [1]. Moreover, the number of small distributed PV generators increases the complexity of the electricity management for the grid operator. The energy transition increases electricity demand, particularly with the new electricity uses as EV, data centres … and requires more renewable and so intermittent power. While this transition places additional stress on power grids, SGs can help manage it and reduce the need for costly new infrastructures improving grid resilience and reliability [7,8]. These fundamental changes in modes of production, distribution and consumption led to the development of smart grids and micro-grids. A new concept has also emerged, the energy community, whose definitions may vary but whose objective is to produce, consume, and share energy generally produced from renewable sources while generating economic, environmental, and social benefits—and in a geographical perimeter, the limits of which depend on the legislation in force. An EC is a virtual construct, with virtual energy flows; thus, ECs and microgrids are two separate concepts that do not relate in general; in an EC Sharing rules can be defined in different ways, often relatively simple to be easily applicable but not always optimal for obtaining the greatest benefit from an economic, environmental, social and fairness point of view. Optimizing these exchanges of virtual or real energy is a real challenge and is related to this article.

Before focusing on ECs, it is useful to recall the distinction between the two main operating modes of individual PV generation. In a full feed-in scheme, all PV generation is injected into the public grid and remunerated according to the applicable feed-in tariff. In a surplus feed-in scheme, PV generation is first consumed on site, thereby reducing the consumer electricity bill, and only the surplus is injected into the public

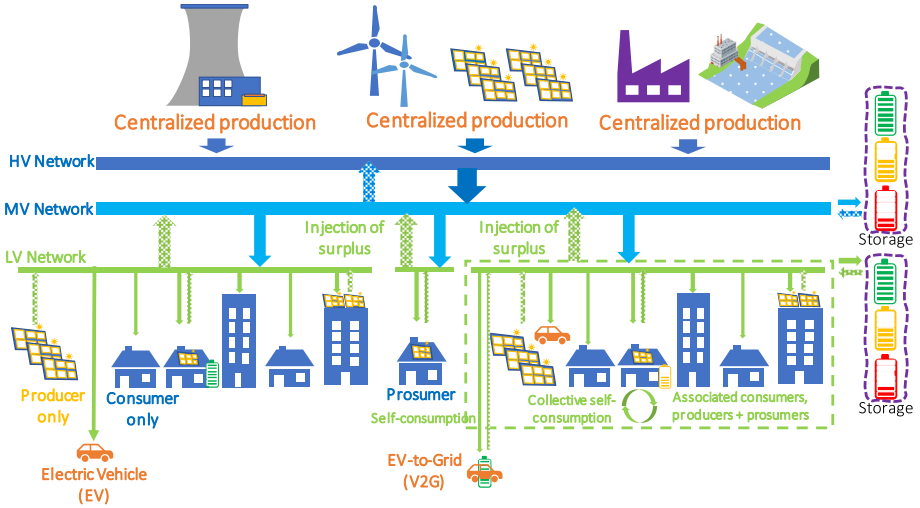


Fig. 1. The future of electricity networks adapted from Ref. [6].

grid and remunerated under a surplus feed-in tariff. Fig. 2 illustrates these two configurations.

For the consumer, PV self-consumption makes it possible to cover part of the electricity demand with local generation and therefore to reduce the electricity bill. At community level, collective self-consumption can support renewable integration and improve the temporal matching between local generation and demand.

SC should play an increasingly important role in its individual or collective form. The use of an ESS (particularly battery) facilitates the production/consumption balance and allows to reduce or avoid PV production losses. The adoption of EVs is also increasing, and they play an important role, particularly as flexible loads that can be utilized during periods of high solar production [9,10]. The development of vehicle-to-grid (V2G) is emerging, based on the principle of bidirectional operation, whereby the grid can draw on the energy stored in vehicle batteries. In other words, V2G offers the ability to charge or discharge the battery of an electric (or hybrid) vehicle depending on demand and network capacity [11].

In an ISC system, the user has an interest in adapting, as much as possible, his consumption with his production. Some loads in a building or a city are shiftable and can be smartly connected and disconnected according to the production in order to match consumption and production [12–14]. Self-consumption (SC) refers to the share of PV production that is directly used to meet the load. However, if an ESS is charged exclusively with PV, the energy discharged from ESS can also be considered as SC [15].

The aim is thus to increase energy performances and to reduce the bill, often by increasing the SC part. Usual dimensionless indices quantify the performance of ISC PV system (Fig. 3) [15,16]:

- Self-consumption rate (SCR), the ratio between the share of the PV energy produced and consumed on site and the total PV production:

$$SCR = \frac{Self\ Consumption\ (SC)}{PV\ Production} \tag{1}$$

SCR can reach 100% but it implies that the PV plant is small compared to the consumption which is rarely interesting.

- Self-Sufficiency Rate (SSR), the ratio between the share of the PV energy produced and consumed on site and total consumption:

$$SSR = \frac{Self\ Consumption\ (SC)}{Consumption} \tag{2}$$

A SSR of 100% means that all the consumption is covered by on-site production which is rarely possible, especially without energy storage.

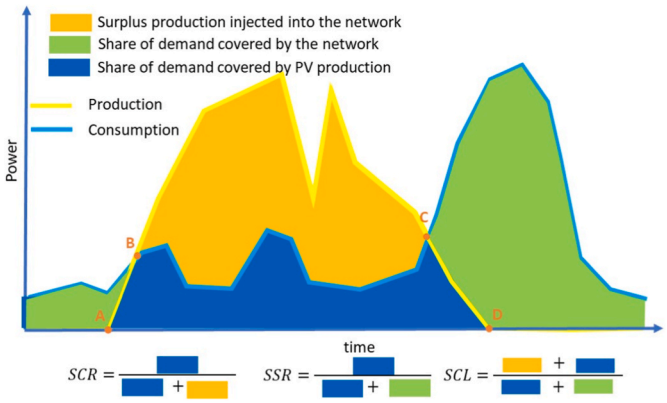


Fig. 3. Illustration of self-consumption and definition of some performance's metrics adapted from Ref. [17].

- Another metric can be used, Solar Coverage Level (SCL), which is the ratio between the total PV production and the total consumption:

$$SCL = \frac{PV\ Production}{Consumption} = \frac{SSR}{SCR} \tag{3}$$

This metric can exceed 100% and is generally used to present monthly or annual energy balances rather than as a performance indicator.

Achieving strong ISC performance would require both a high SCR and a high SSR, although these two indicators tend to exhibit opposite behaviours [18]. To have a high SCR, the share of consumption synchronized with PV production must be high and the PV installed power too. To have a high SSR, all the PV production must be consumed which can be achieved by reducing the PV installed power (the small amount of energy produced is then guaranteed to be consumed).

To increase these performance metrics, a solution consists in shifting the loads (such as water heating and EV charging) towards the hours with high solar production or to store the electricity surplus in a ESS for later use. However, the investment on a battery storage, even small in size, must be economically justified. The Li-ion battery cell price is decreasing significantly (by 73% in 10 years) with an average price falling from 290 $/kWh in July 2014 to 77.82 $/kWh in July 2024 [19,20] and estimated at 74$/kWh in 2025 by BloombergNEF [21]. The battery pack, for its part, is estimated at $108 per kWh [21], although it remains not yet widely deployed in these production systems; only 2% of ISC installations are combined with ESS in France [22] in March 2025. Sharing electricity with several people, especially with different consumption profiles, is a potential solution called Collective

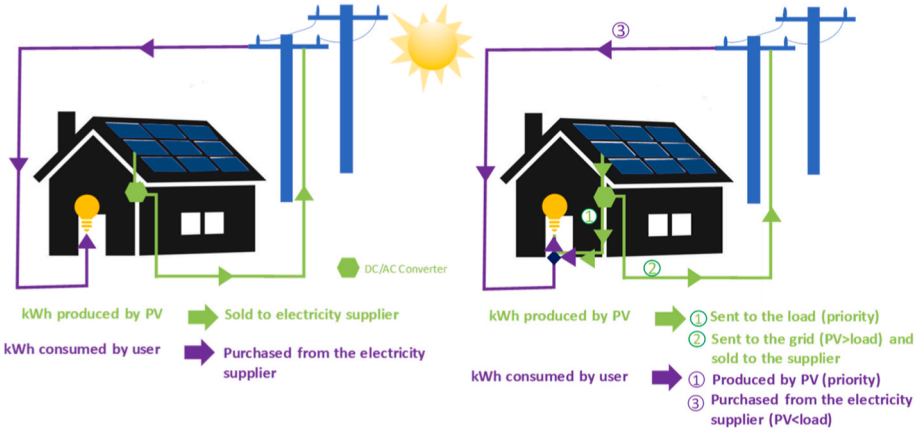


Fig. 2. The two types of individual PV installation: Total sale where all the PV electricity produced is sold to the electricity manager (Utilities) (left drawing) – SC with the sale of surplus (right drawing).

Self-Consumption (CSC). It enables electricity exchanges between geographically close producers and consumers, connected through clearly defined sharing rules. This increases SCR through greater local coordination of production and consumption. If the energy produced is entirely self-consumed at the network level, it is not injected into the network of a higher voltage domain and therefore has no impact.

CSC is now possible thanks to technological advances, particularly digital technology [23,24]. The development of smart meters has enabled the real time transmission of data (usually in 15- or 30-min time steps). If compared to numerous actors involved in ISC, CSC has a major advantage, which is expressed in the different load profiles. It is well known that the main obstacle to overcome in order to support large-scale RES integration is matching demand with production in real time [22,25] (Fig. 4).

SCR and SSR were extended for ISC with storage [15], considering that SC consists of two terms: PV production directly supplying the load $PV_{\rightarrow load}$ and passing by ESS before supplying the load $PV_{\rightarrow ESS}$.

$$SCR_{ESS} = \frac{PV_{\rightarrow load} + ESS_{\rightarrow load}}{PV\ Production} = \frac{PV_{\rightarrow load} + ESS_{\rightarrow load}}{PV_{\rightarrow load} + PV_{\rightarrow grid} + PV_{\rightarrow ESS}} \quad (3)$$

$$SSR_{ESS} = \frac{PV_{\rightarrow load} + ESS_{\rightarrow load}}{PV_{\rightarrow load} + ESS_{\rightarrow load} + GRID_{\rightarrow load}} \quad (4)$$

Branchetti et al. [16,26] modified the expressions of SCR and SSR for CSC in adding in the SC the Shared Energy (SE) defined as the energy virtually exchanged within the perimeter of the EC, defined at each time step by:

$$SE = \min\left(GRID_{\rightarrow CSC}; PV_{CSC\rightarrow grid}\right) \quad (5)$$

$GRID_{\rightarrow CSC}$ being the energy withdrawn by CSC from the grid and $PV_{CSC\rightarrow grid}$ the energy produced in the CSC and fed into the grid. Then,

$$SCR_{CSC} = \frac{PV_{CSC} - PV_{\rightarrow grid}}{CSC\ PV\ Production} \quad (6)$$

$$SSR_{CSC} = \frac{PV_{CSC} - PV_{CSC\rightarrow grid}}{CSC\ Consumption} \quad (7)$$

PV_{CSC} is the PV energy produced by all the CSC and $PV_{CSC\rightarrow grid}$, the share of this production fed to the grid.

It is evident that the performance of CSCs depends on several factors, including the composition of the EC (complementarity of load profiles), the production potential (solar resources and installed capacities), the rules governing ESh within the EC and with the main grid as well as the objectives pursued by the community (objective function for optimization).

The regulations and rules relating to the establishment of EC vary from country to country and some of them are presented in Ref. [27]. ESh is most often done through distribution keys which correspond to coefficients that allocate the production of CSC operation among the consumers who participate. The method for calculating these coefficients, or the coefficients themselves are provided at each time step to the distribution network manager [28]. However, it is important to understand the difference between the physical and economic aspects. In the CSCs context, energy injected into the network, or energy withdrawn from the network refers to the SC at the contractual accounting level rather than the physical reality.

The choice of a distribution key [29] is flexible and has a significant impact on the economic outcome for a given physical configuration. The economic outcome of such a CSC system depends strongly on the level of synchronisation between consumption and production [30] and on how benefits are distributed among prosumers.

The scope of this review is limited to electricity-based energy communities. Sector coupling and heat energy communities represent important related topics, particularly for local decarbonization strategies, but they involve different infrastructures, physical constraints, tariff structures, and regulatory frameworks. They are therefore acknowledged here as relevant extensions, but they are not included in the analytical framework developed in this paper [31].

The topic of EC is broad and has been the subject of numerous research and review articles. An assessment of reviews published over the last five years (58 papers) was carried out to position the present contribution with respect to existing syntheses (Table 1). The objective of this review-of-reviews step is not to provide a bibliometric analysis, but to identify which dimensions of EC research have already been extensively reviewed and which aspects remain less systematically synthesized, particularly energy-sharing mechanisms, optimization-based coordination, and internal tariff design.

This paper adopts a structured analytical review approach. It does not claim to be a PRISMA-based systematic review or a bibliometric study. Instead, it combines a review-of-reviews positioning step with a focused analytical synthesis of energy-sharing allocation rules, optimization-based coordination, distributed and market-based mechanisms, hybrid architectures, and internal tariff structures.

The review-of-reviews step was conducted using keyword searches

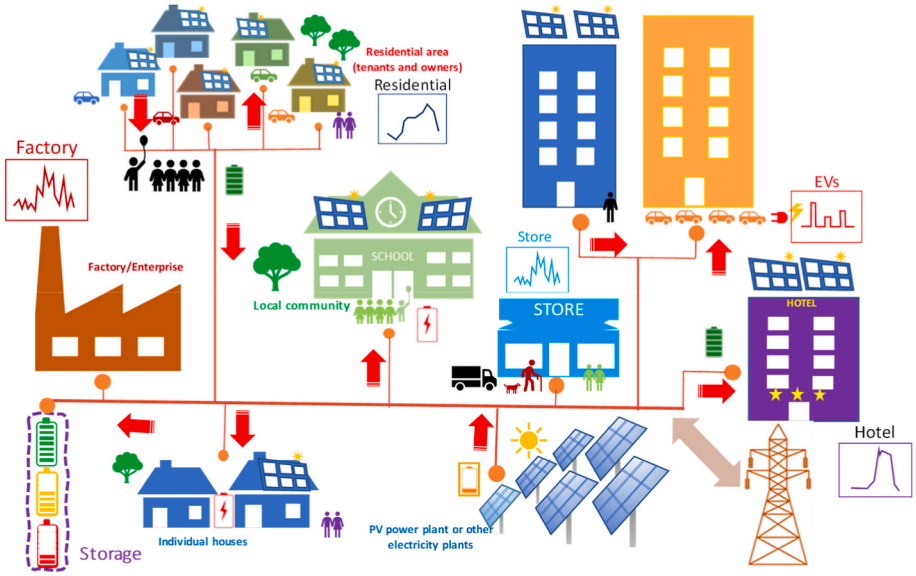


Fig. 4. Illustration of electricity exchange in CSC and complementarity of loads (the consumption profiles are plotted in the boxes).

combining terms such as “energy community”, “collective self-consumption”, “renewable energy community”, “energy sharing”, “peer-to-peer energy”, “optimization”, “tariff”, “pricing”, “fairness”, and “review”. The purpose was to identify recent review papers and determine whether the specific interaction between sharing rules, optimization methods, and internal tariff structures had already been comprehensively addressed.

The focused synthesis then selected studies according to their relevance to the analytical dimensions of this paper: allocation rules, welfare-oriented optimization, distributed decomposition, market-based or equilibrium models, storage coordination, fairness, and tariff design. Studies exclusively devoted to legal definitions, social acceptance, generic business models, or non-electric energy communities were considered for positioning when relevant, but were not included in the detailed analytical framework unless they contributed directly to energy-sharing design.

About 15% of the reviews present a general overview of EC, 40% deal with business policy and social aspects and the rest concerns various technical and engineering sides. Only 9 reviews address at least one of the topics covered in our article (in bold and blue in Table 1) and we point out these dissimilarities below.

Li et al. [80] comprehensively proposed different methods for

Table 1
Summary of review articles published over the last five years on energy communities – In bold and blue, the themes related to those covered in this article [32–79].

Ref N°	Year	Contents	Comments & Main conclusions	Area	Nb of papers cited
[32]	2020	Elements characterizing EC, ECs worldwide, EC strategy development.	The concept of EC is clarified; it is not just an exchange of electricity but also of social and collective interest exchanges. The well-being of the population is at the centre of the optimization of territorial planning; the legislative and regulatory context essential for an orderly management and without risks of the energy system.	General	112
[33]	2021	Bibliometric study using 6 databases and keywords, Concepts and overlapping of definitions creating confusion.	Definition of EC is different according to the authors: The most commonly used term is ICES (Integrated Community Energy System); Research must be developed on synergic improvement in multi-purpose EC; the most underlined benefit of EC is the decrease of CO2 emissions and the increase in efficiency and self-sufficiency; Many papers are about existing policies, there is not radical reform of energy governance inducing some difficulty to overcome political barriers.	Research trends	103
[34]	2021	EC topologies, EC terminology, EC categories: most of ECs uses PV and ESS, P2P trading.	As noted in ref [32] EC terminology showed that some terms are well-defined, while others are simply different names for the same types of EC; Most ECs use PV, P2P trading is an essential part of EC; Microgrids appear in only 16% of publications, which may suggest that they are not technologically necessary for achieving energy savings; using aggregators is a good approach to make the development of EC easier.	EC Structure	76
[35]	2021	EC comparisons in terms of physical and cost structure, **Cost allocation approaches, Tariff design objectives**, Cost allocation procedures and regulatory principles.	Cost allocation in EC is a major factor in its success, and this topic is quite new and relatively understudied.	Cost	123
[36]	2021	EC business models;	8 business model archetypes identified for EC to identify the market; most of EC are in SC mode with surplus generation trading; the initial shared investment and the provision of the complete value chain are the main reasons explaining why EC are the most widely exploited collective initiatives.	Business, Policy	104
[37]	2021	Policies for EC development, Feed-in-Tariffs to support EC projects.	EC projects create a multitude of benefits while being driven by a wide range of interests. This makes it difficult for policy makers to design policies that live up to all the different agendas that actors might have.	Business, Policy	140
[38]	2021	Policies for EC development, international regulatory framework, Italian case.	The review used the Glasziou’s approach, using both scientific papers, regulatory documents, and case study reports, and considering the main drivers for the current transformation of power systems into SG. EC experiences are growing all around the world, showing the great interest of National Governments, Research Institutes, private companies, and end-users for this new model. Future perspectives of increasing liberalization and access of different actors to the electricity market suggest that technologies like blockchain if widely adopted can provide a unique language for all parties	Business, Policy	102
[39]	2021	Assessment, comparison and discussion about the approaches to national transposition of the EU provisions for EC.	The status of transposition of the EU provisions is diverse and represents different degrees of ambition and fitting into the national context; the developed frameworks will need to materialise as attractive business models for the involved actors.	Business, Policy	55
[40]	2022	Microgrids from controllable energy system to interactive communication bridge, Blockchain-enabled decentralized framework for multi-scale communication, Application surveys, challenges and future trends.	The co-creation of microgrid and blockchain will enhance the self-organization of the grid edge and promote the cutting-edge innovations by effectively interacting different stakeholders from academic researchers, industrial partners and decision makers.	Control, Communication	119
[41]	2022	Common characteristics of EC, Fundamental scopes, Principal evaluation indexes.	Novel indicators must be developed or already existent ones must be borrowed from other fields of research for their measurement	General	111
[42]	2022	Regional energy internet, Complex network topology modelling, Optimal network flow allocation.	This paper synthetizes the particularity of coupling interconnection between transportation network and social network in regional energy internet; The concept of energy self-organization (ESO) and EC is introduced based on self-organization theory, meanwhile, the differences and advantages between ESO and previous similar concepts are summarized and compared.	Network, Transmission	173
[43]	2022	Reviewing of numerous articles to clarify the notion of EC terms.	183 definitions of ECs were found in 405 articles; the meanings lied to the notion of EC and the objectives pursued by EC vary substantially across concepts and over time; there is a growing focus on communities’ economic objectives rather than their social or political goals	General	124
[44]	2022	Market models, Characteristics of P2P, EC and Transactive Energy markets.	P2P, CSC and transactive energy markets were compared in term of definition, participation, governance structure, topology, and designs. There are five evidence gaps requiring future research before these markets could be widely adopted which are the lack of: consideration of physical constraints; a holistic approach to market design and operation; consideration about how these market designs will scale; consideration of information security; and, consideration of market participant privacy.	Business, Policy	175

[45]	2022	Factors for the emergence of ECs, Bibliometric analysis, Content analysis, Geographic analysis.	Generic factors have contributed to the emergence of ECs regardless of their location, such as appropriate policies and support for practitioners. Other factors, such as the geography of the transition and cognitive-cultural factors, are less studied	Business, Policy	96
[46]	2022	Business models of EC, EC network topologies, Future EC network topologies, Analysis of 3 case studies in Netherlands and Denmark.	All the projects analysed proved that the participation of EC members is crucial to succeed. There are different triggers of social acceptance and willingness to participate that play a key role in the adoption process. The implementation of digital technology makes sense only if participants are eager to share their data. The replication of the projects in another location is very limited, thus all location-specific aspects should be carefully examined.	Business, Policy, Topology	91
[47]	2022	Standalone communication technologies, Communication standards, Technologies and issues/challenges.	This paper provides insights into communication standards and technologies, issues/challenges, and future research perspectives for Interoperable Smart Microgrids implementation. Advanced wireless technology, called LoRa (Long Range), and a modern architecture using it to establish a communication network is shown.	Network, Transmission	132
[29]	2023	Regulations of EC, Review of sharing criteria, Performance ratio, Case studies.	The criteria for allocating the electricity produced among the shareholders are analyzed (in the Spanish context) and evaluated (static and dynamic allocation coefficients) by observing their performance and applicability, and by classifying them in order to maximize the savings of the self-consuming participants. the highest savings occur when electricity is allocated following distribution coefficients considering the customer's consumption better than investment participation or contracted power.	EC rules	38
[48]	2022	General overview	No methods exist to assist decision-maker, not decision-making tools, such as optimization models, to support decision-maker with low knowledge in RE integration through the actual building design process.	General	76
[49]	2023	Developments in demand-side ESh and CSC schemes, Definition and category of sharing economy, ESh technologies and some future directions.	Classification of sharing economy are presented, with a focus on the applications in the energy sector: virtual power plants, P2P energy trading, shared energy storage, and microgrid ESh cloud. An outlook on demand-side ESh technologies is presented.	Energy Management	105
[50]	2023	ESh rules, Factors influencing shared consumption, Bibliometric study, Barriers.	The review shows that the last 3 years have seen a significant surge in EC research output and reveals the need to extend research beyond specific sectors such as accommodation and transportation.	EC sharing rules	222
[51]	2023	Roles and interaction of stakeholders, Relevant mechanisms for designing flexibility provisions, Flexible energy markets for local energy communities, Key barriers for energy community implementation.	This paper reviews designing flexibility provisions for local ECs in the context of existing and emerging flexible electricity markets and discusses the need for social arrangements, technical designs, and their impact on EC.	Economics	140
[52]	2023	Bibliometric analysis, Analysis of interactions of prosumers and utilities	Several interesting points are discussed: Analysis of prosumers and utility interactions with the moment of user energy consumption, the time of exporting energy to the grid by the prosumers, the time of consumer feeding or storage, the moment for managing the stored energy considering the best price	EC sharing rules	216
[53]	2023	EC actors, European level regulations	This study reviews the EC regulations at European and national levels for Austria, Bulgaria, Germany, Ireland, Latvia, Poland, and Spain and shows that RES production targets and participation are considered more important than the economic benefits that could be induced by ECs.	Business, Policy	162
[54]	2023	Review of P2P, Community SC and transactive energy models	This paper examines business models of actors operating in local energy markets; For each type of market actor, a business model archetype is determined, characterised using the business model canvas and discussed.	Business, Policy	184
[55]	2023	Sharing and trading, evolution of private laws for EC	The review highlighted that there is not a clear understanding of what exactly distinguishes trading from sharing of electricity and this distinction has relevant consequences in Law, including for what concerns a value-oriented analysis between individualism based and pluralism-based models	Business, Policy	55
[56]	2023	Energy modelling, Business models, Objectives in EC, Objective function and optimisation methods	A focus on the inclusion and a description of business models and objective functions is realized. The paper highlights significant research opportunities at the crossroad among technical design, policy, and economics, especially when considering multi-energy and sector-coupled system. The objective functions retrieved in the selected articles were grouped into 4-categories depending on the focus of the optimisation: economic, environmental, technical, and social. Optimisation algorithms were classified into 3 classes: exact algorithms, heuristic algorithms, and simulator.	ESh, Optimisation	171
[57]	2023	Classification of EC, Number and repartition of EC in Europa	The authors found nearly 4000 energy communities with 900,000 members in the European Union. The situation of ECs is analysed: energy sources used, existing associations in each country, number of ECs, ... more information on existing ECs would be needed.	Business, Policy	82

[58]	2023	Situation of EC in Italy, legislative framework in EU	Interviews and surveys were collected with market operators, both on the service providers' side (ESCo) and the communities' side (promoters), to identify conflicts and advantages associated with establishing communities	Business, Policy	97
[59]	2024	Community-based Virtual Power Plant, Circular economy, Definition and presentation	This study presents a CE-based cVPP model, that combines the principles of circular economy (CE) and community-based virtual power plant (cVPP). The authors investigate the enduring and sustainable benefits of the CE-cVPP across four dimensions: energy, environment, economics, and society.	General	158
[60]	2024	Local energy markets in competitive energy market, Large-scale and local energy markets, Algorithms for energy transactions.	This paper reviews the different energy trading projects initiated at the global level and machine learning approaches and solution strategies for local energy markets. Reinforcement learning algorithms are classified into model-free and model-based methods considering the agent type, learning methods, policy, state space, action space, and action selection for state, action, and reward function outputs.	Economics	255
[61]	2024	Role of materiality and its relation to collective agency, Inclusion, Approaches to participation, Types of community innovations through various methods and disciplines	The objectives of the paper are to better understand how researchers understand and approach materiality and material engagement in community energy innovation.	Social	114
[62]	2024	Very general information not only for EC	The paper analyses the merits and demerits of each energy production system, discuss the utilization of various energy storages and it classifies systematic optimization into 3 categories based on different application scenarios: building level, neighbourhood level, and community level.	General	160
[63]	2024	Characteristics, Trends, Business models and Legal framework for EC	This article covers almost all aspects of ECs, allowing for a general overview of the current state of science and technology in this field.	Business, Policy	146
[64]	2024	Carbon credit, Carbon emissions reduction potential by EC	The study explores whether carbon credits could serve as a practical tool for funding the development of ECs. The paper looks 10 EC projects in Europe and shows that there is potential for using carbon credits to finance energy community initiatives	Policy, Environment	127
[65]	2024	Bibliometric study, Energy retrofit technologies	This review focuses on technological advancements in building energy retrofits within the context of smart and connected communities (S&CC).	Building	200
[66]	2024	Decentralised energy trading models, Characterisation of P2P, Transactive energy and EC	The objective of this paper is to determine the key defining characteristics of energy systems labelled as P2P, Community/CSC or Transactive Energy. This study is based on the results of 5 expert group interviews, representing a wide range sectoral perspectives from 13 countries and an analysis of 78 academic and 55 non-academic papers.	Business	129
[67]	2024	General overview	It is very general review on renewable EC and their concepts, benefits, types, technological components relevant to energy efficiency measures, energy storage and energy monitoring.	General	228
[68]	2024	Overview of Italian regulatory framework for ECs and identification of challenges and opportunities for EC	This review overviews the Italian regulatory framework for ECs and identifies challenges and opportunities for ECs development in terms of system operation, finance, and economics.	Business, Policy	62
[69]	2025	EC forms of aggregation and diffusion in the world, Importance of buildings and their integration into new power grid, Frameworks, Case studies analysis.	This review shows the classification of aggregation forms and the worldwide development of ECs. The paper also explores the integration of various components (RES, ESS and smart technologies). It presents demand response strategies applications and shows the importance of prosumers behaviour to achieve energy effectiveness. Case studies are also analysed to show the key features in EC scenarios.	General, Bibliometric study management	150
[70]	2025	Conventional **Optimization-Based solutions** (with objective functions), strategies and tools to implement in EC, Demand Response, forecasting, and load management goals, **ML-based solution methods**.	The objective of this review is to ensure that Machine Learning (ML) based solutions are essential tools to study and further discussed the several ML-based algorithms based on their objectives. It compares the possible strategies and tools that could be implemented in EC. Different solution tools are analysed based on their advantages, such as Demand Response (DR), forecasting, and load management goals.	Energy management	148
[71]	2025	Implementation of EU regulatory frameworks for ECs in Germany, Spain and Italy, Main barriers and driving factors influencing operation and management.	This paper reviews how Germany, Spain, and Italy implemented the Renewable Energy Directive II and the Internal Energy Market Directive by identifying their operational and management impacts on ECs. The authors showed how national policy differences shape ECs development and evaluated the determining factors and barriers influencing the development of ECs.	Business, Policy	164
[72]	2025	Social benefits, Social and institutional perspectives	This review shows that technical perspectives dominate the EC research field, while social and institutional perspectives are less prominent. and often neglected in research. The authors are questioned if promotion of ECs will, by necessity, reduce energy poverty, increase social equity and hamper energy segregation.	Social Aspect	40

[73]	2025	Policy and challenges in EU, Netherlands, Germany, Austria, Italy, Greece, Denmark, Sweden, Norway, Finland, UK	The situation of ECS in several European countries is analysed. The challenges often stem from limited national support and difficulties in fully achieving diversity within engaged local ECs. It is recommended to build on early learnings in EC and further strength local anchoring to achieve a just transition.	Business, Policy	107
[74]	2025	Benefits of EC, Impact on the grid, Load shedding, grid charges	An optimization model is developed to allocate the demand and production of each participant, generation unit, and storage initially belonging to different communities that are implemented under the same distribution transformer and engage in multiple participation. Both energy- and grid-related costs are minimized.	Economy	63
[75]	2025	Fairness in EC, Fairness indexes, Centralized and decentralized EC management, ESh	This work studies the Pool market and P2P transactions to investigate the impact of different bidding from the users' perspective. The 10 centralized and decentralized approaches are assessed based on economic performances at both users' and community levels. Specific attention is paid to fairness within the community. 3 indexes derived from economy and game theory are considered. Results show that the pool market returns considerable savings among decentralized frameworks compared to P2P markets.	Social, ESh	47
[76]	2025	Challenges and need for resilient energy systems including engineering, computer science, hazards geography, technoeconomic analysis and public policy.	This paper concerns a transdisciplinary approach to evaluating the challenges and need for resilient energy systems that include engineering, computer science, hazards geography, technoeconomic analysis, and public policy. It reviews relevant work from these various disciplines and proposes a transdisciplinary, geospatial framework focused on equitable grid resilience.	General	183
[77]	2025	Bibliometric study	This paper reviews the existing literature on this EC topic of growing scholarly interest. The findings highlight a lack of homogeneity in the literature in conceptualising the benefits of renewable energy communities for different stakeholders.	Research trends	133
[78]	2025	Socio-cultural factor, Bibliometric study, Economic aspect, Technological aspect, Legal aspect for Latin America	The authors conduct a comparative analysis of critical elements in the development of ECs in Latin America and developed nations, with the objective of discerning challenges and advancements within each region.	Social, economy	87
[79]	2025	EC in social sciences	A bibliometric analysis and review were realized only on social science publications, including geography, sociology, management, economics, jurisprudence and political science.	Social	244
[80]	2025	Transmission planning and scaling of RES in EC, Transmission lines and substations.	The first objective of this paper is reviewing optimal strategies for the integration of RES in ECs and to assess the expansion of transmission infrastructure required to support this integration. ECs play an important role to facilitate the RES integration, the efficiency of this integration is improved through advanced planning methodologies and supportive regulatory frameworks. ECs reduce the need for long-distance transmission through local generation, but the expansion of transmission infrastructure remains essential during periods of high demand.	Electrical networks	175
[81]	2025	Bibliometric study	This study synthesizes critical trends in EC policy and assesses their urgency, in order to understand the prevailing patterns and their ramifications for future studies, stakeholders and policymakers.	Research trends	107
[82]	2025	Definition and characterization of EC, Analysis of technologies and typologies for EC.	A review on recent developments in financial incentives in ECs, their organization, typologies and benefits shared among the participants.	General	101
[83]	2025	Energy storage in EC, management of ESS, Economic consequences of ESS integration	This work addresses the economic incentives for the operation of ECs with centralized community storage or stationery and mobile storages. The operating equations for storage systems in various situations are given. The complexity of EC operation with ESS is shown, and a significant interdependence is demonstrated between participants, different ESSs, price structures and price developments when considering variable retail prices.	Economics	59
[84]	2026	Current state of EC regulation in the Member States (cooperation between DSO and ECs), analysis of the Spanish decree for ECs.	More effort is needed for a better cooperation between the DSO and ECs. A lack of innovative technological equipment and solutions to facilitate the exchange of information is noted. There is no linear relationship between the number of ECs per capita and the gross domestic product per capita of a country or the number of legal forms that can become an EC.	Legislative	111
[85]	2026	Distributed-optimization methods for demand response, storage scheduling and distributed-generation dispatch, performance limits & claimed benefits, blockchain, artificial intelligence and game-theoretic coordination	This review article contains the most information on the topics covered in this article. This paper synthesizes distributed optimization as a practical management paradigm for ECs, identifies key application areas and profiles scalability, privacy, and resilience characteristics. The main technological levers are reviewed: blockchain/distributed ledgers, artificial intelligence, and game theory concepts are analysed to observe how they promote secure data exchange, real-time coordination, and incentive compatibility within multi-agent energy networks. The challenges related to distributed optimization at the scale of electricity are studied, including convergence in the presence of heterogeneity, time-varying conditions, communication delays, cybersecurity, and confidentiality guarantees.	Optimization, ESh	193
[86]	2026	Structure, Internal organization and management, Trading and financial management, Support incentives.	This review analyses the operational characteristics of ECs, control strategies, market interactions, regulatory frameworks, and real-world implementation challenges.	Economic	213
[87]	2026	Socio-economic interrelationships in West Africa.	This review looks the socio-economic consequences of microgrid PV systems in and on vulnerable West African communities, with a focus on the food-water-energy and health-energy nexus.	Social, Economy	81
[88]	2026	Urban energy modelling.	This paper investigates the integration of renewable ECs in large urban districts, using a GIS-based approach to evaluate the aggregation strategies and their effects on energy and economic performance.	GIS	81

designing tariffs but did not assess their impact on EC benefits. Llera-Sastresa et al. [29] present sharing criteria and their consequences on economic performance, but limited only to European regulatory frameworks. Zhu et al. [81] describe the different sharing methods but do not elaborate on their scope. Khalek and Chakraborty [82] conducted a bibliometric study on sharing rules without going into detail about the content of the publications. Barabino et al. [83] determined the occurrences of the different optimization methods without in-depth description or critical analysis. Asim Amin et al. [84] present a large part of the objective functions and optimization methods, particularly AI, but describe them succinctly without evaluating their performance. Gasca et al. [85] explain the principles of profit redistribution comprehensively, incorporating the notion of fairness, but they disregard the optimization of energy flows within ECs. The review by Basaran et al. [86] is the one that most closely matches our objectives; however, the authors address numerous aspects, which leads them to provide only a concise treatment of optimization methods and their effectiveness.

Optimizing energy exchanges is central to improving EC performance because allocation rules, coordination mechanisms, and internal tariffs jointly determine technical indicators, aggregate welfare,

individual savings, and fairness. For this reason, the present review focuses on ESh methods within CSC schemes, mathematical approaches for optimizing power flows, pricing models between EC members, and the associated economic and distributive outcomes. The mechanisms discussed are not limited to currently implemented regulatory arrangements; technically feasible approaches are also considered when they clarify the design space of EC coordination.

The preceding review of recent review articles shows that EC research is distributed across several domains, including energy systems, electrical engineering, economics, regulation, sociology, and policy. The present paper does not aim to provide another broad overview of ECs. Its contribution is a structured analytical review of energy-sharing mechanisms, optimization-based coordination methods, and internal tariff structures. It examines how these dimensions interact and how they affect efficiency, incentive compatibility, scalability, fairness, and regulatory feasibility. The structure of the paper is shown in Fig. 5.

Section 2 constitutes the analytical core of the article. It first establishes a unified physical and operational framework for ESh within EC (Section 2.1), defining the common feasibility space that underlies all coordination mechanisms. A centralized social planner benchmark is then introduced (Section 2.2) as a normative welfare reference. Section 2.3 clarifies the different notions of efficiency used in the literature. Building on this foundation, Section 2.4 interprets rule-based, allocation-based, distributed, market-based, and hybrid mechanisms as alternative solution concepts applied to the same constrained optimization problem. Sections 2.5 to 2.8 provide a structured comparative analysis of these paradigms along dimensions of optimality, incentives, scalability, fairness, and regulatory embedding. Section 2.9 highlights tariffs design and financial outcomes efficiency among the different ESh mechanisms.

Section 3 discusses the practical implications of the unified framework for EC design, regulation, tariff implementation, and comparative assessment.

Section 4 synthesizes the main research gaps identified across coordination paradigms and discusses emerging trends, including integrated mechanism design, hybrid multi-layer architectures, adaptive allocation rules, and regulatory feasibility challenges. Section 5 concludes by summarizing the analytical contributions of the review and outlining implications for the design and evaluation of energy-sharing architectures.

2. Review on energy sharing and optimization

Before reviewing individual energy-sharing mechanisms, this section first introduces the analytical lens used throughout the paper. The objective is not to propose a new operational optimization model, but to clarify how apparently different coordination mechanisms can be compared when they are interpreted as alternative ways of organizing decisions over the same physical energy system.

The section therefore proceeds in four steps. First, a common physical feasibility framework is introduced, based on energy balance, grid exchanges, and storage dynamics. Second, a centralized social planner formulation is used as a benchmark for welfare-oriented coordination. Third, the main notions of efficiency used in the literature are disentangled, since technical self-consumption indicators, aggregate welfare, and individual savings do not measure the same property. Fourth, rule-based, centralized, distributed, market-based, and hybrid mechanisms are reviewed as different coordination logics applied to this common constrained system.

This structure is intended to make explicit that the reviewed mechanisms differ primarily in the way information, decision authority, price formation, and fairness considerations are organized, rather than in the underlying physical system itself.

2.1. A unified physical and optimization framework for energy sharing

ESh mechanisms within EC can be interpreted as alternative coordination solutions defined over a common underlying physical system. Although regulatory frameworks, institutional governance, and economic settlement rules differ substantially across implementations, all sharing architectures operate under identical physical constraints: nodal power balance, intertemporal storage dynamics, and grid interaction limits. Establishing this unified analytical foundation is essential to enable rigorous comparison between rule-based (RB), allocation-based, market-based, and hybrid paradigms beyond purely procedural descriptions.

Consider a community composed of N participants indexed by $i \in N$, over a discrete time horizon $t \in T$. For each participant, let:

- $E_i^{gen}(t)$ denotes local renewable generation,
- $E_i^{load}(t)$ denotes electricity demand,
- $E_i^{grid}(t)$ denotes net exchange with the public grid (positive when importing),
- $E_{ij}(t)$ denotes bilateral exchanges within the community.

The nodal energy balance constraint is:

$$E_i^{gen}(t) - E_i^{load}(t) + E_i^{grid}(t) + \sum_{j \neq i} E_{ij}(t) = 0 \quad \forall i, t \tag{8}$$

At the community level:

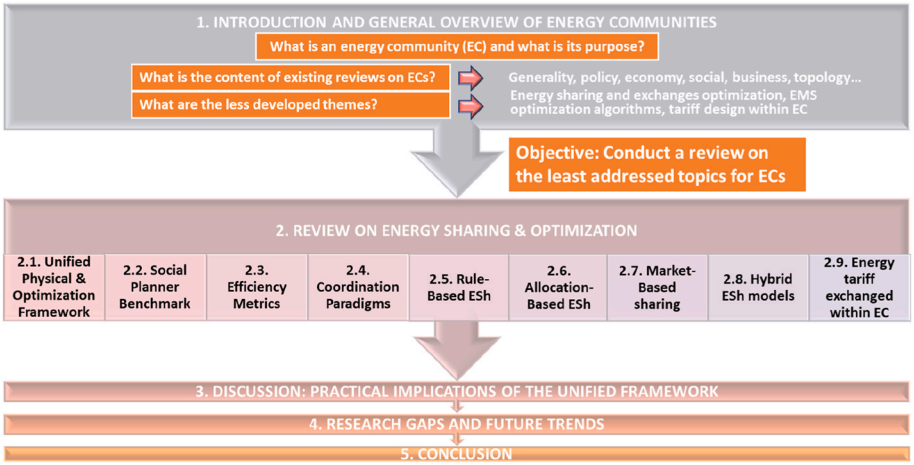


Fig. 5. Structure of the review.

$$\sum_{i \in N} E_i^{grid}(t) = E^{community.net}(t) \tag{9}$$

When shared ESS is present, intertemporal feasibility requires:

$$SOC(t+1) = SOC(t) + \eta_c E^{ch}(t) - \frac{1}{\eta_d} E^{dis}(t) \tag{10}$$

subject to capacity bounds:

$$SOC^{min} \leq SOC(t) \leq SOC^{max} \tag{11}$$

These constraints define the feasible operational space common to all ESh mechanisms, independently of the coordination logic used to determine allocations or prices.

In practical terms, this means that all coordination mechanisms considered in this review must solve the same basic physical problem: local generation, demand, storage operation, internal exchanges, and grid imports or exports must be made mutually consistent at each time step. What changes from one paradigm to another is not the physical feasibility requirement itself, but the rule used to decide who receives energy, who pays whom, how prices are formed, and whether fairness or incentive constraints are explicitly considered.

This distinction is important because many studies compare energy-sharing schemes through their final outcomes without making clear whether observed differences come from the physical system, the allocation rule, the price-setting procedure, or the regulatory settlement mechanism. The unified framework introduced here separates these layers and therefore provides a common basis for comparing otherwise heterogeneous studies.

Within this space, different sharing paradigms can be understood as distinct solution concepts applied to a shared constrained optimization structure. Centralized allocation directly solves a welfare-maximization program over this feasible set. Distributed approaches approximate that solution through decomposition algorithms. Market-based mechanisms induce decentralized equilibria subject to the same coupling constraints. RB schemes restrict the feasible allocation space ex ante through predefined coefficients. Hybrid models articulate combinations of these logics across multiple coordination layers.

This unifying perspective allows comparison across mechanisms in terms of optimality guarantees, computational scalability, incentive compatibility, fairness properties, and regulatory embedding dimensions that are often treated separately in the literature on CSC and P2P energy systems [85,87–89]. Fig. 6 synthesizes the unified physical framework underlying all ESh mechanisms.

The diagram emphasizes that nodal balance constraints, intertemporal storage dynamics, and grid interaction limits define a common operational space. The subsequent coordination paradigms discussed in Sections 2.5–2.8 differ not in physical structure, but in the institutional and mathematical logic applied to solve this shared constrained system.

Having established the common physical feasibility space, the next step is to define a benchmark against which alternative coordination mechanisms can be compared. This benchmark is provided by the centralized social planner formulation.

2.2. Social planner benchmark

The social planner benchmark is used here as an analytical reference, not necessarily as a governance prescription. It represents the outcome that would be obtained if a coordinating entity had full information about participants' generation, demand, flexibility, storage constraints, and costs, and could choose the community-wide allocation that minimizes total cost or maximizes aggregate surplus.

This benchmark is useful because it provides a reference point against which other mechanisms can be interpreted. A rule-based mechanism may be administratively simple but deviate from this benchmark; a distributed algorithm may approximate it under convexity and truthful participation; a market-based mechanism may reproduce it only under restrictive assumptions on competition and price-taking behaviour.

A natural analytical benchmark for ESh consists of a centralized welfare-maximization problem defined over the feasible operational space introduced above. This formulation provides a normative reference against which alternative coordination paradigms can be evaluated.

Let $x_i(t)$ denote the decision variables associated with participant i, including controllable demand, storage operation, and grid exchanges. A generic centralized formulation writes:

$$\min_{x_i} \left(\sum_{i \in N} C_i(x_i) + C_{grid}\left(P^{community.net}\right) \right) \tag{12}$$

subject to:

- nodal power balance constraints,
- storage intertemporal dynamics,
- device-level feasibility sets $x_i \in X_i$.

This formulation encompasses cost-minimization models widely studied in CSC and EC literature [87,90,91]. Extensions incorporate demand response, shared storage coordination, tariff interactions, and fairness-aware objectives.

Under convexity assumptions, the centralized problem admits well-defined optimality conditions, and the associated Lagrange multipliers provide economically meaningful signals representing the marginal system value of electricity. This dual interpretation establishes a

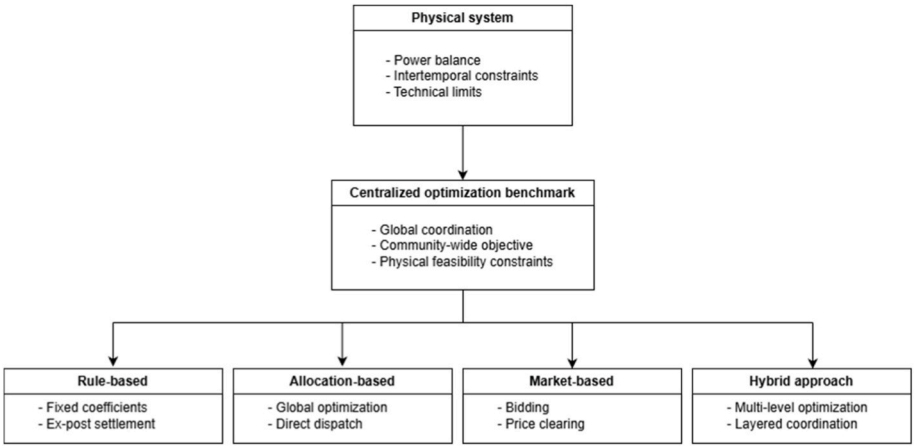


Fig. 6. Flowchart of the unified physical framework.

theoretical bridge to pricing-based P2P markets, where decentralized agents respond to internal prices instead of a centrally imposed allocation.

Importantly, the centralized formulation does not merely represent an implementable governance structure. Rather, it serves as a normative efficiency benchmark. Deviations observed under RB, distributed, or market-based mechanisms can therefore be interpreted as resulting from informational decentralization, strategic behaviour, regulatory embedding, or computational constraints.

Positioning the centralized planner as a benchmark clarifies that differences across ESh paradigms are not rooted in distinct physical systems, but in alternative coordination logics applied to a common constrained optimization problem.

The social planner benchmark defines one notion of optimality, but it does not by itself resolve how performance should be measured. The next subsection therefore clarifies the different meanings of efficiency used in the energy community literature.

2.3. Defining and disentangling efficiency metrics

In this paper, efficiency is not treated as a single metric. Three levels must be distinguished. Technical efficiency refers to physical indicators such as self-consumption rate, self-sufficiency rate, or grid imports. Allocative efficiency refers to the maximization of aggregate community welfare or minimization of total system cost. Individual economic efficiency refers to the savings obtained by each participant relative to a reference case without energy sharing.

These three dimensions may move in different directions. A mechanism can increase collective self-consumption while producing an unequal distribution of benefits. Conversely, a fairness-constrained allocation may improve individual acceptability while reducing aggregate welfare. For this reason, the review explicitly distinguishes the metric being considered when comparing coordination mechanisms.

The term “efficiency” is frequently invoked in the EC literature, yet it encompasses distinct and sometimes conflicting notions. Comparative assessment of ESh mechanisms requires explicit identification of the metric under consideration, as different coordination paradigms may perform differently depending on the evaluation criterion. Two primary categories of efficiency indicators can be distinguished.

2.3.1. Allocative efficiency (welfare maximization)

Allocative efficiency refers to the maximization of aggregate surplus within the community:

$$W = \sum_{i \in N} U_i(x_i) - C_{grid} \tag{13}$$

where U_i denotes participant utility (or negative cost). This definition corresponds to the objective of the social planner benchmark and serves as the normative efficiency reference in optimization-based studies [87,88].

Allocative efficiency captures the total economic benefit generated by coordinated operation but does not provide information about its distribution across participants.

2.3.2. Individual cost-based metrics

From a participant-level perspective, performance is frequently measured through individual cost reductions:

$$\Delta C_i = C_i^{baseline} - C_i^{community} \tag{14}$$

Such metrics are central in empirical evaluations of sharing schemes and strongly influence adoption decisions. However, maximizing aggregate surplus does not guarantee that all participants experience positive individual gains, particularly in heterogeneous communities.

2.3.3. Conceptual implications

Failure to distinguish between these metrics can lead to misleading comparisons across coordination paradigms. For example:

- A rule-based scheme may achieve high SCR (see Eq. (6)) while remaining allocatively inefficient.
- A market-based equilibrium may maximize aggregate surplus but produce unfair benefit distribution.
- A fairness-constrained centralized solution may sacrifice total welfare to improve distributive equity.

It is therefore necessary to specify explicitly which notion of efficiency is invoked when evaluating ESh mechanisms.

Once these efficiency notions are distinguished, the different families of energy-sharing mechanisms can be interpreted as alternative ways of organizing decisions over the same feasible system. This is the purpose of the following subsection.

2.4. Coordination paradigms as alternative solution concepts

Within the unified framework introduced above, ESh mechanisms can be interpreted as alternative solution concepts applied to a common constrained optimization problem. The physical system and feasibility set remain identical across paradigms; what differs is the coordination logic used to determine allocations and prices. This perspective is consistent with welfare-based formulations of CSC and community ESS coordination developed in Refs. [87,90,91], as well as equilibrium-based formulations of P2P electricity markets in Ref. [88].

RB sharing restricts the feasible allocation space ex ante through predefined coefficients, thereby reducing the decision problem to a constrained redistribution rule. Empirical and regulatory analyses of such coefficient-based mechanisms are discussed in Refs. [29,92–95], which highlight how allocation rules modify economic outcomes without altering the underlying physical constraints.

Centralized allocation directly solves the social planner's optimization program, yielding welfare-optimal dispatch under full information [87,90,91,96]. Distributed allocation decomposes this centralized problem into coordinated local subproblems, typically through dual decomposition or augmented Lagrangian methods [97–99], approximating the centralized optimum while relaxing data centralization requirements. Under convexity assumptions, these distributed schemes preserve the welfare-theoretic structure of the centralized benchmark.

Market-based sharing defines decentralized agent optimization problems whose interaction produces equilibrium allocations. Such mechanisms are commonly formalized as generalized Nash equilibrium (GNE) or auction-clearing models [88,100–102]. In these architectures, allocations emerge from strategic interaction rather than direct welfare maximization, even though equivalence with the centralized solution may arise under restrictive assumptions [88].

Hybrid models combine centralized feasibility enforcement with decentralized price-mediated interaction across multiple coordination layers, as illustrated in transactive energy and hybrid trading schemes [85,103]. These architectures explicitly separate physical dispatch from economic settlement while preserving a shared feasible operational space.

Interpreting these paradigms as distinct solution concepts enables systematic comparison along analytically meaningful dimensions, including allocative efficiency [87,88], computational scalability [97,98], incentive compatibility [88,100], fairness properties [85], and regulatory compatibility [92,93]. Rather than representing fundamentally different physical systems, these mechanisms correspond to different institutional and mathematical approaches to solving the same underlying constrained coordination problem.

The following subsections examine these coordination paradigms in increasing order of institutional and computational complexity, starting with rule-based allocation and progressively moving toward centralized,

distributed, market-based, and hybrid architectures.

2.5. RB energy sharing

RB ESh constitutes the most institutionally widespread coordination mechanism in CSC frameworks, particularly within European regulatory environments. Its defining feature is the use of predefined allocation coefficients that determine how locally generated energy is distributed among community members, independently of real-time optimization.

2.5.1. Formal structure of sharing coefficients

Let $\alpha_i(t)$ denote the allocation coefficient assigned to participant i at time t, satisfying:

$$\alpha_i(t) \geq 0, \sum_{i \in N} \alpha_i(t) = 1, \forall t \tag{15}$$

If total community generation at time t is:

$$E_{tot}^{gen}(t) = \sum_{i \in N} E_i^{gen}(t) \tag{16}$$

then the shared energy allocated to participant i is:

$$E_i^{share}(t) = \alpha_i(t) \cdot E_{tot}^{gen}(t) \tag{17}$$

This formulation reflects the standard regulatory definition of CSC allocation rules in several European frameworks [29,92,93]. In practice, coefficients may be:

- static ($\alpha_i(t) = \alpha_i$),
- periodically updated,
- or computed according to predefined proportional rules (e.g., contracted capacity, historical consumption).

Although this structure appears neutral, the choice of coefficient rule implicitly defines the distribution of economic benefits and the degree of temporal alignment between generation and demand. Empirical analyses show that alternative coefficient definitions can significantly alter both CSC ratios and individual bill savings [94,95].

From an optimization perspective, RB sharing restricts the allocation space to a one-dimensional proportional redistribution at each time step. Unlike centralized dispatch models where allocations respond endogenously to marginal costs [90,91,96], coefficient-based schemes decouple allocation from short-term system state variables. This structural decoupling explains why RB mechanisms may perform sub-optimally when generation and load profiles are heterogeneous across participants.

2.5.2. Regulatory embedding and virtual net billing

Virtual Net Billing (VNB) schemes represent a canonical implementation of RB sharing. Under VNB, the distribution system operator (DSO) records individual production and consumption and applies allocation coefficients ex post for settlement purposes [93,95].

Let $P_i^{grid,+}(t)$ and $P_i^{grid,-}(t)$ denote, respectively, the electricity imported from and exported to the public grid by participant i at time t. Given the shared energy allocation $E_i^{share}(t)$, the electricity bill of participant i over the time horizon t can be expressed as:

$$B_i = \sum_{t \in T} \left[\pi^{buy}(t) E_i^{grid,+}(t) - \pi^{sell}(t) E_i^{grid,-}(t) + \pi^{comm} E_i^{share}(t) \right] \tag{18}$$

where:

- $\pi^{buy}(t)$ the retail import tariff (per kWh),
- $\pi^{sell}(t)$ the export remuneration tariff (per kWh),
- $\pi^{comm}(t)$ the internal community transfer price associated with shared generation (per kWh).

The first term captures the cost of net imports from the grid, the second term reflects revenues from individual surplus injections, and the third term represents the internal economic settlement associated with the allocated share of community generation.

Although VNB-type mechanisms are administratively simple and regulator-compatible, their economic performance depends critically on the predefined sharing coefficients $\alpha_i(t)$, which remain decoupled from real-time demand heterogeneity. As a result, allocation does not necessarily coincide with the welfare-maximizing dispatch derived from centralized optimization.

2.5.3. Efficiency limitations

From the unified optimization perspective introduced earlier, RB ESh restricts the feasible allocation space to:

$$E_i^{share}(t) = \alpha_i(t) E_{tot}^{gen}(t) \tag{19}$$

which generally differs from the welfare-maximizing allocation obtained under centralized coordination.

When temporal mismatch exists between generation and individual demand, static coefficients may allocate energy to participants whose consumption does not coincide with production, thereby increasing grid injections and reducing CSC [96,104]. This effect is particularly pronounced in communities with heterogeneous load patterns or unequal PV ownership.

Formally, if x^{RB} denotes the RB allocation and x^* the centralized optimum:

$$W(x^{RB}) \leq W(x^*) \tag{20}$$

with strict inequality in the presence of demand heterogeneity, storage flexibility, or non-linear tariff structures.

Beyond allocative efficiency, RB mechanisms may also generate distortions in grid cost recovery and tariff redistribution. Fina [97] shows that alternative redistribution mechanisms can significantly alter participant incentives and grid fee fairness, even when aggregate self-consumption remains unchanged. Similarly, Gasca et al. [85] demonstrate that centralized and decentralized coordination frameworks differ substantially in their ability to balance efficiency and equity objectives.

Importantly, these inefficiencies are not merely computational but structural. Unlike distributed optimization methods that approximate the centralized optimum under convexity [86], RB schemes do not incorporate marginal system costs into allocation decisions. Consequently, they cannot internalize intertemporal storage arbitrage or demand response flexibility in an endogenous manner [90,105].

Therefore, RB sharing does not guarantee allocative efficiency and may induce persistent welfare losses relative to optimization-based coordination.

2.5.4. Dynamic and optimized coefficients

To mitigate the structural inefficiencies of static proportional rules, recent research has reframed coefficient design as an optimization problem rather than a purely administrative parameter choice.

Instead of computing a fixed α_i , coefficients can be computed dynamically to minimize grid exchanges, maximize collective self-consumption, or improve distributive fairness:

$$\min_{\alpha_i(t)} C\left(P^{community,net}(t)\right) \tag{21}$$

subject to feasibility constraints on $\alpha_i(t)$ and regulatory admissibility.

Gianaroli et al. [106] introduce dynamic sharing keys designed to adapt to temporal variability in generation and load, demonstrating measurable improvements in collective performance compared to static schemes. Similarly, Villalonga Palou et al. [107] formulate coefficient estimation as a meta-heuristic optimization problem, highlighting the sensitivity of economic outcomes to the coefficient selection process.

Salazar-Hamze and Guerrero-Lemus [108] propose an optimization-based framework tailored to Spanish energy communities, showing that coefficient recalibration can significantly reduce grid dependency.

More recently, Madrigal et al. [104] employ machine learning techniques (XGBoost-based prediction) to anticipate optimal allocation coefficients under demand variability. This data-driven approach illustrates a shift from rule prescription toward adaptive allocation mechanisms.

From a fairness perspective, Jorquera-Bravo et al. [109] argues that coefficient design should incorporate explicit equity objectives rather than relying solely on proportional allocation. Embedding fairness constraints at the coefficient level partially bridges the gap between RB systems and cooperative allocation frameworks.

However, even dynamically optimized coefficients remain structurally distinct from centralized dispatch. They adjust redistribution parameters ex ante but do not endogenously determine real-time energy flows through marginal cost signals. Consequently, dynamic coefficient optimization reduces—but does not eliminate—the allocative gap relative to welfare-maximizing coordination.

The Austrian context provides a particularly relevant example of the practical distinction between static and dynamic allocation mechanisms. In such frameworks, static coefficients provide administrative simplicity and predictability, whereas dynamic allocation keys can better reflect time-varying consumption and generation patterns. The comparison between these approaches illustrates that allocation coefficients should not be treated as neutral settlement parameters: they directly affect self-consumption indicators, individual savings, and fairness among participants. This supports the broader view adopted in this review, according to which rule-based sharing mechanisms must be evaluated not only in terms of regulatory simplicity, but also in terms of their allocative and distributive consequences [110,111].

2.5.5. Fairness and implicit redistribution

Although RB systems lack endogenous price formation, they embed implicit redistributive effects. The allocation of surplus:

$$S_i = C_i^{baseline} - C_i^{community} \tag{22}$$

depends directly on the coefficient structure.

Static proportional rules may concentrate benefits among participants with favourable load-generation coincidence, larger installed capacity, or higher contracted power, thereby raising equity concerns [97,112,113]. Empirical evidence suggests that redistribution mechanisms significantly influence perceived fairness and long-term participation incentives within EC [29,92].

To assess distributive properties more formally, cooperative game-theoretic allocation rules are often used as normative benchmarks. Let $v(S)$ denote the value generated by coalition $S \subseteq N$. The Shapley value assigns to participant i the average marginal contribution across all coalition orderings:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|} [v(S \cup \{i\}) - v(S)] \tag{23}$$

This allocation rule satisfies efficiency, symmetry, additivity, and guarantees that the total surplus is fully distributed [87,114]. However, its computation becomes combinatorically intensive as community size increases.

Alternative cooperative fairness criteria include the nucleolus, which minimizes the maximum dissatisfaction (excess) across all coalitions by solving a lexicographic minimization problem. This approach prioritizes coalition stability by reducing incentives for subgroup deviation. Gasca et al. [85] compare centralized and decentralized fairness formulations and show that incorporating cooperative benchmarks can significantly alter dispatch and redistribution outcomes.

Another widely used criterion is envy-freeness, which requires that no participant prefers another's allocation to their own. While conceptually appealing, envy-freeness is rarely satisfied under simple proportional sharing rules unless coefficients are specifically calibrated to participant heterogeneity [109,113].

Consequently, RB CSC mechanisms do not inherently guarantee coalition stability, envy-freeness, or proportional fairness. Their distributive properties are entirely determined by the chosen coefficient rule, which may reflect regulatory simplicity rather than equity optimization.

2.6. Allocation-based energy sharing

Allocation-based mechanisms determine the distribution of locally generated energy by explicitly solving an optimization problem over a defined time horizon. Unlike RB schemes, allocations are not predetermined but emerge from a coordinated decision-making process that internalizes physical, economic, and regulatory constraints.

Within the unified framework introduced earlier, allocation-based sharing corresponds to solving a welfare-oriented optimization problem directly over the feasible operational space.

2.6.1. Centralized allocation framework

In centralized allocation architectures, a single coordinating entity (e.g., community operator or aggregator) determines energy exchanges by solving a global optimization problem over the full-time horizon. This paradigm constitutes the normative benchmark for collective self-consumption systems and has been extensively studied in both residential and multi-energy contexts [90,91,105].

Let $x_i(t)$ denote the controllable decision variables of participant i at time t, including flexible demand, storage operation, and grid exchanges. A generic centralized formulation writes:

$$\min_{x_i(t)} \left(\sum_{t \in T} \left[\sum_{i \in N} C_i(x_i(t)) + C_{grid}\left(P^{community,net}(t)\right) \right] \right) \tag{24}$$

subject to power balance and device-level feasibility constraints.

In [105] this framework is extended to multi-energy microgrids, demonstrating how joint optimization of sizing and operational management improves system-wide efficiency. A two-stage management strategy integrating operational dispatch and billing mechanisms is proposed in Ref. [90], illustrating the interaction between physical optimization and economic settlement. Similarly, Gil Mena et al. [91] analyse centralized CSC in Spanish residential buildings and quantify the gains relative to static sharing rules.

Unlike RB mechanisms, centralized allocation internalizes marginal system costs and intertemporal flexibility. When community ESS is present, dispatch decisions are coupled through state-of-charge dynamics, enabling temporal arbitrage and peak shaving [87,114,115].

2.6.2. Economic interpretation via dual variables

Under convexity and regularity conditions, the centralized allocation problem admits KKT optimality conditions (i.e. Lagrangian gradient is equal to 0). Beyond their mathematical role, these conditions provide a physically meaningful interpretation of coordination signals within constrained energy systems.

The role of dual variables can be interpreted without requiring the reader to follow the full optimization derivation. When a constraint is binding, its associated dual variable measures the marginal value of relaxing that constraint. In the present context, the multiplier associated with the community power balance can be interpreted as an internal marginal value of electricity at a given time step. This value is not necessarily a market price, but it provides a theoretical bridge between centralized optimization and price-based coordination mechanisms.

The Lagrangian $\mathscr{L}$ consists in the total system operating costs, plus for each time step a term representing the penalty associated with any imbalance in the power balance equation, and additional terms

enforcing temporal dynamics and physical device constraints.

Explicitly written,

$$\mathscr{L}(\mathrm{x},\lambda,\mu)=\sum_{t,i}C_i(x_i(t))+\sum_t\lambda_t r_t+\sum_t\mu_t h_t \tag{25}$$

where:

- r_t denotes the residual of the power balance at time t;
- h_t represents the temporal and device-level constraints (e.g., storage dynamics);
- λ_t is the multiplier associated with the power balance constraint;
- μ_t are the multipliers associated with the dynamic constraints.

The multiplier λ_t^*, the multiplier associated with the optimum value, admits an economic interpretation as the marginal internal value of electricity at time t. Under convexity and price-taking assumptions, decentralized decisions responding to this signal reproduce the centralized optimum, thereby establishing the theoretical bridge between welfare maximization and price-based peer-to-peer coordination [88].

2.6.3. Demand response and demand-side management

A first major extension of centralized allocation concerns the integration of demand response (DR) and demand-side management (DSM) strategies. These mechanisms aim to temporally reshape electricity consumption in order to better align demand with local renewable generation.

As discussed by Yang et al. [87] centralized coordination allows flexible loads to be scheduled jointly with PV generation, thereby increasing CSC and reducing grid exchanges. DR typically refers to short-term reactive adjustments to price or control signals, whereas DSM encompasses longer-term structural modifications of consumption patterns [115].

Optimization-based DSM formulations often include shiftable or interruptible loads characterized by duration constraints and fixed power requirements. Centralized scheduling of such loads has been shown to smooth aggregate demand profiles and enhance local renewable utilization. However, computational complexity increases rapidly with the number of controllable devices, making large-scale implementations sensitive to problem size and motivating partially distributed alternatives [91].

2.6.4. Community ESS integration

ESS represents a second major extension of centralized allocation. By enabling temporal shifting of surplus PV production, shared ESS enhances both SC and operational flexibility.

Centralized coordination of CES is typically formulated through intertemporal state-of-charge dynamics (Eq. (3)) and is subject to capacity bounds (Eq. (4)).

It has been demonstrated that coordinated operation of shared storage generally yields higher collective benefits than individual battery ownership [87,115]. However, the distribution of ESS-related gains becomes a secondary allocation problem. Coalition stability depends critically on the rule used to distribute costs and benefits among participants.

Moreover, the interaction between storage and tariff design plays a decisive role. Dynamic pricing structures may unintentionally increase peak imports if not coupled with capacity-based components, highlighting that ESS optimization cannot be evaluated independently of the regulatory environment [115].

Recent work on economic incentives for different storage concepts further shows that the value of community storage depends not only on its technical operation, but also on the ownership structure, the internal settlement rule, and the price signals faced by participants. Centralized community storage, individually owned stationary storage, and mobile storage resources may generate different incentives and distributive effects. Consequently, storage integration should be analysed jointly with tariff design and benefit-allocation rules rather than treated as a purely technical flexibility resource [116].

2.6.5. Fairness-constrained optimization

While centralized coordination maximizes aggregate welfare, it does not guarantee equitable surplus distribution. Several works incorporate fairness objectives directly into the optimization problem.

Jorquera-Bravo et al. [109] formulate fair energy allocation in CSC frameworks by embedding equity constraints within the dispatch problem. Gasca et al. [85] compare centralized and decentralized fairness models and show that introducing inequality penalties can significantly modify dispatch outcomes. Fina [97] further demonstrates that grid cost redistribution mechanisms alter participant incentives even when physical dispatch remains unchanged.

A generalized fairness-constrained problem can be expressed as:

$$\min_x\left(\sum_i C_i(x_i)+\gamma\cdot\Phi(S_1,\ldots,S_N)\right) \tag{26}$$

where Φ represents an inequality metric over participant surpluses S_i, and γ controls the efficiency–equity trade-off.

Gasca et al. [85] provide a systematic comparison of centralized and decentralized fairness frameworks in energy communities, showing that explicit fairness constraints can significantly alter dispatch outcomes. Similarly, cooperative allocation methods such as the Shapley value have been proposed to ensure coalition stability and prevent subgroup deviation [87].

These developments illustrate that centralized allocation has evolved from pure cost minimization toward multi-objective coordination frameworks integrating efficiency, stability, and distributive fairness.

2.6.6. Distributed optimization and decomposition methods

As the size and heterogeneity of EC increase, fully centralized optimization may become computationally burdensome and institutionally impractical. The requirement to collect detailed consumption, generation, and flexibility data from all participants also raises concerns regarding privacy, communication overhead, and governance centralization.

Distributed allocation mechanisms address these limitations by decomposing the centralized welfare maximization problem into coordinated local subproblems. Instead of solving a single monolithic optimization program, participants (or local agents) iteratively solve individual decision problems while exchanging coordination signals.

A canonical approach relies on dual decomposition. Introducing the Lagrange multiplier $\lambda(t)$ associated with the community power balance constraint:

$$\sum_{i\in N}x_i(t)=0 \tag{27}$$

the centralized problem can be decomposed into participant-level subproblems of the form:

$$\min_{x_i(t)\in X_i}\left(C_i(x_i(t))+\lambda(t)x_i(t)\right) \tag{28}$$

The coordinating entity updates $\lambda(t)$ iteratively to enforce feasibility. Under convexity assumptions, convergence to the centralized optimum can be guaranteed.

To enhance convergence stability, many works employ augmented Lagrangian approaches such as the Alternating Direction Method of Multipliers (ADMM). Mbuwir et al. [98] demonstrate distributed scheduling of community microgrids using consensus optimization, showing convergence toward welfare-optimal solutions without full data sharing. Cheng et al. [117] extend distributed coordination over longer optimization horizons within energy internet architectures.

Liu et al. [99] apply ADMM to networked microgrids embedded in distribution systems, highlighting robustness under communication constraints. Khojasteh et al. [118] proposes a robust ADMM-based framework explicitly tailored to EC, incorporating uncertainty and operational risk.

More recently, López et al. [119] introduce stochastic profile steering to coordinate distributed EC members, while Ahmadi et al. [120] propose a user-centric decentralized multi-objective management system balancing economic and environmental objectives.

Basaran et al. [86] provide a comprehensive survey of distributed optimization methods in sustainable energy communities, confirming that ADMM-based architectures dominate large-scale implementations due to scalability and privacy preservation.

2.6.7. Scalability and information decentralization

Distributed optimization preserves the welfare-theoretic structure of centralized allocation while relaxing data centralization requirements. Participants retain control over private cost functions and device constraints, sharing only limited coordination variables.

This architecture improves scalability and institutional acceptability, particularly in large communities where computational burden and data governance concerns become significant [91]. It also aligns with regulatory contexts where full operational centralization may not be legally or socially desirable.

However, distributed allocation remains fundamentally different from market-based coordination. Agents in distributed optimization typically behave as price-takers responding to algorithmic signals rather than as strategic actors maximizing individual surplus under endogenous price formation.

Despite theoretical equivalence under convexity, distributed allocation faces several structural limitations:

1. **Nonconvexities** (binary appliance scheduling, integer storage constraints) may break convergence guarantees.
2. **Strategic behaviour** is not explicitly modelled; truthful participation is assumed.
3. **Communication delays and packet loss** may affect convergence stability (Liu et al., 2023).
4. **Regulatory constraints** may prevent implementation of internal dual-based pricing signals.

Zhou and Lund [121] emphasize that distributed coordination without endogenous price formation cannot fully replicate incentive compatibility mechanisms present in peer-to-peer trading markets.

Thus, distributed allocation occupies an intermediate position: it preserves welfare optimality under ideal assumptions while improving scalability and privacy, yet it does not internalize strategic behaviour.

2.6.8. Limitations of distributed welfare coordination

Despite its advantages, distributed allocation retains several structural limitations.

First, convergence guarantees to rely on convexity and well-behaved cost functions. Discrete decisions (e.g., unit commitment, binary appliance scheduling) may break theoretical equivalence with the centralized optimum.

Second, distributed optimization presumes cooperative participation and truthful parameter sharing. Unlike market-based mechanisms, it does not explicitly model strategic behaviour, price manipulation, or withholding incentives.

Third, regulatory embedding may constrain the admissible coordination signals. In some collective self-consumption frameworks, internal prices derived from dual variables cannot be directly implemented due to tariff caps or settlement rules.

Consequently, distributed allocation represents an intermediate coordination architecture: it preserves welfare optimality under ideal assumptions while improving scalability, yet it does not provide endogenous incentive compatibility in the presence of strategic participants.

Distributed allocation occupies a middle position between centralized welfare optimization and fully decentralized market-based coordination. It improves computational tractability and privacy preservation while maintaining normative efficiency under convexity, but it abstracts strategic market dynamics that characterize peer-to-peer trading systems.

2.7. Market-based sharing

2.7.1. Market-based coordination principles

Market-based mechanisms depart from welfare optimization by modelling participants as autonomous agents maximizing individual surplus under endogenous price formation. Rather than solving a centralized program, allocations emerge from strategic interaction.

P2P energy trading architectures have been extensively reviewed by Sousa et al. [100], Soto et al. [101] and Tushar et al. [89], who classify market designs according to pricing rules, clearing mechanisms, and governance structure. Morstyn et al. [102] conceptualize P2P platforms as federated virtual power plants, emphasizing incentive alignment and decentralized coordination.

Formally, such systems can be represented as equilibrium problems in which each agent solves:

$$\max_{x_i \in X_i} (U_i(x_i), p) \tag{29}$$

subject to coupling constraints linking all participants. The resulting allocation corresponds to a GNE when shared constraints (e.g., power balance, line limits) are present.

Le Cadre et al. [88] provide a rigorous formulation of peer-to-peer electricity trading as a variational inequality (VI) and GNE equilibrium problem. Under convexity and price-taking assumptions, they show that decentralized equilibrium allocations may coincide with the centralized welfare optimum when equilibrium conditions replicate the KKT conditions of the planner's problem.

Wörner et al. [122] highlight that real-world P2P implementations rarely satisfy ideal competitive assumptions, and user value propositions depend heavily on pricing architecture.

2.7.2. Non-priced matching mechanisms

Not all P2P mechanisms rely on endogenous price formation. Matching-based systems determine bilateral trades according to predefined rules, priority rankings, or geographic constraints.

Let $M \subseteq N \times N$ denote admissible matches. A generic allocation problem can be expressed as:

$$\max_{M} \left(\sum_{\{(i,j) \in M} w_{ij} \right) \tag{30}$$

subject to feasibility constraints such as line capacity and topology.

Such mechanisms are computationally tractable and operationally scalable. Geographically constrained matching has been shown to reduce congestion and enhance local reliability [123]. However, because no explicit marginal price coordinates behaviour, welfare guarantees and incentive compatibility cannot be derived from equilibrium theory.

Empirical analyses indicate that economic benefits under matching-based P2P depend strongly on tariff design and technology penetration [124]. Consequently, efficiency is typically assessed ex post through aggregate metrics rather than derived from equilibrium properties.

2.7.3. Endogenous price formation and auction mechanisms

Fully transactive P2P markets introduce explicit price formation through equilibrium computation, auction-based clearing, or distributed negotiation protocols.

In double-auction formulations, participants submit bids b_i and asks a_j. A clearing price p^* satisfies:

$$\sum_{i \in N_{buy}} q_i(p^*) = \sum_{j \in N_{sell}} q_j(p^*) \tag{31}$$

Under competitive assumptions, uniform-price double auctions approximate marginal cost pricing [125,126]. Variants integrating blockchain infrastructure or trust-aware mechanisms have been proposed to enhance transparency and mitigate manipulation risks [127, 128].

Comparative studies show that clearing-rule design significantly affects surplus distribution, convergence stability, and price volatility [129]. Thus, endogenous pricing mechanisms are not homogeneous: their strategic and distributive properties depend strongly on architectural design choices.

2.7.4. Strategic behaviour and efficiency limits

Unlike distributed welfare optimization, market-based mechanisms explicitly model strategic participants. Agents may engage in:

- price manipulation,
- withholding strategies,
- asymmetric information exploitation.

As a result, equilibrium efficiency does not automatically imply social optimality. The divergence between equilibrium allocation x^{eq} and welfare optimum x^* can be measured through welfare loss metrics such as the Price of Anarchy [88].

Furthermore, fairness concerns emerge as a structural dimension. Irusta and Morales [130] incorporate Wasserstein-distance constraints into equilibrium-based trading to bound distributional inequality. The Wasserstein distance measures the minimal transport cost required to transform one surplus distribution into another; constraining it bounds inequality between participants. Their formulation transforms equilibrium computation into a constrained mechanism design problem integrating efficiency and equity.

Market-based systems therefore offer endogenous marginal price signals capable of aligning private incentives with system objectives under restrictive assumptions. However, they introduce strategic complexity, computational overhead, and regulatory oversight requirements absent in pure allocation-based models.

Market-based ESh occupies the high-complexity end of the coordination spectrum. It internalizes incentive compatibility and investment signalling but trades off simplicity and guaranteed welfare optimality for strategic decentralization.

2.8. Hybrid energy sharing models

2.8.1. Hybrid coordination as multi-layer architecture

Hybrid ESh architectures combine centralized optimization and decentralized price-mediated interaction within structured multi-layer coordination systems. They do not constitute a third equilibrium concept per se, but rather an institutional and mathematical articulation between welfare-based allocation and market-based coordination [85, 103].

Khorasany et al. [131] propose a hybrid trading scheme in transactive energy markets where centralized feasibility enforcement coexists with peer-level price negotiation. This dual-layer architecture reflects a broader trend: separating physical dispatch from economic settlement.

Hybrid systems typically operate across two interacting layers:

- A technical layer, ensuring physical feasibility (power balance, network constraints, storage dynamics);
- An economic layer, governing pricing, transfers, and settlement mechanisms.

Let x denote physical dispatch decisions and p internal prices. Hybrid models determine (x, p) jointly, but through distinct coordination logics at each layer.

2.8.2. Allocation-first architectures

In allocation-first hybrids, a centralized optimization problem is solved first to determine technically feasible dispatch:

$$x^* = \underset{x}{\operatorname{argmin}} C(x) \tag{32}$$

Economic transfers are subsequently computed using dual variables or predefined redistribution rules.

Let $\lambda^*(t)$ denote the Lagrange multiplier associated with the power balance constraint. Internal prices are then defined as:

$$p^{dual}(t) = \lambda^*(t) \tag{33}$$

Under convexity, $p^{dual}(t)$ represents the marginal system value of electricity [88]. However, Moreno et al. [95] and Gallego-Castillo et al. [94] show that regulatory constraints in European CSC frameworks often restrict admissible internal pricing schemes. As a result, dual-based prices may not be directly implementable within legal settlement rules.

Fina [96] further demonstrates that grid cost redistribution mechanisms significantly influence participant incentives even when physical dispatch remains welfare-optimal. This creates a structural separation between technical optimality and regulatory feasibility.

2.8.3. Market-first architectures

Market-first hybrids reverse the sequencing logic. Participants first engage in decentralized trading governed by endogenous price formation:

$$x^{eq} = GNE(p) \tag{34}$$

after which a supervisory entity enforces feasibility corrections:

$$x^{corr} = \Pi_F(x^{eq}) \tag{35}$$

where Π_F denotes projection onto the feasible physical set.

Distributed coordination schemes such as ADMM are frequently used to reconcile decentralized decisions with network constraints [118, 132]. In this configuration:

- Marginal price consistency is prioritized,
- Scalability is enhanced through decentralized computation,
- Feasibility enforcement is externalized to a supervisory authority.

However, Gasca et al. [85] emphasize that fairness constraints embedded post-clearing may alter equilibrium incentives. This reveals a tension between stability, efficiency, and equity in hybrid coordination.

2.8.4. Dual–equilibrium price divergence

A central analytical tension in hybrid architectures concerns the potential divergence between:

$p^{dual}(t)$: shadow price from centralized optimization,

$p^{eq}(t)$: decentralized market-clearing price.

Under convexity and price-taking behaviour:

$$p^{dual}(t) = p^{eq}(t) \tag{36}$$

In practice, divergence arises due to:

- nonconvex decision variables,
- strategic bidding,
- tariff caps,
- fairness constraints,
- regulatory settlement rules.

Zhou and Lund [121] argue that pricing models critically shape

decision-making behaviour in smart communities, reinforcing that dual–equilibrium consistency cannot be assumed outside theoretical benchmarks.

As seen in Section 2.7.4, Irusta et Morales [130] used the Wasserstein-distance to incorporate fairness constraints within P2P trading. Embedding such constraints into hybrid architectures modifies equilibrium conditions and may further widen dual–equilibrium gaps.

2.8.5. Regulatory embedding and governance constraints

Hybrid systems operate within legally defined frameworks governing collective self-consumption, tariff structures, and grid fee redistribution [94,95].

Regulatory constraints restrict admissible pricing sets:

$$p(t) \in P^{reg} \tag{37}$$

Such embedding often breaks theoretical equivalence between dual prices and equilibrium prices. Many optimization-based models treat regulation as an exogenous constraint rather than an endogenous design parameter, limiting implementation realism.

Hybrid systems therefore require coordination not only across optimization layers but also across governance authorities, including DSOs, community operators, aggregators, and regulators.

2.8.6. Cross-layer fairness challenges

Let $S_i(x,p)$ denote participant surplus money. Hybrid systems may impose distributive constraints of the form:

$$D(S) \leq \delta \tag{38}$$

where D is an inequality metric.

However, embedding fairness simultaneously at both dispatch and pricing layers remains an open research problem [85]. Existing studies typically optimize efficiency at one layer while addressing equity through ex post redistribution.

Hybrid ESh represents an architectural synthesis rather than a distinct paradigm. It attempts to balance:

- welfare efficiency,
- incentive compatibility,
- scalability,
- regulatory compliance.

However, unresolved tensions between shadow-price optimality, equilibrium incentives, and governance constraints define the current research frontier [85,97].

Table 2 synthesizes the analytical properties of the main coordination paradigms discussed above. Rather than providing a descriptive taxonomy, the comparison highlights structural trade-offs between efficiency, strategic compatibility, scalability, and regulatory embedding.

2.9. Internal tariffs and financial outcomes in energy communities

While previous sections focused on the methods used to distribute energy within ECs and the mathematical methods to optimize this sharing, energy allocation alone does not determine the economic outcomes. The financial performance of such schemes ultimately depends on how electricity is priced, how network costs are incorporated, and how tariffs interact with allocation rules. This section focuses on tariff design and its impact on EC performance. It specifically examines how electricity tariffs are defined within EC, how they influence economic outcomes, and how different calculation methods address efficiency and fairness in the distribution of costs and benefits among participants.

Six categories of business models have been applied depending on national and local regulation [83]. The applicability of twelve cost allocation mechanisms to EC, for tariff design in large power systems, has been discussed [80], providing useful insights; 7 out of 12 mechanisms could be applicable in this context, with varying levels of ease of implementation. Mello et al. [133] proposed a post-delivery pool market based on a fair local market design approach. Depending on the pricing mechanism used, the price of electricity can be known in advance or only after the electricity is delivered. With the “bill sharing” method, the total EC energy bill is shared among its participants, depending on their energy consumption and generation. All members pay and receive the same price for consumed and exported energy. With the “Mid-Market Rate”, the internal energy price is simply the average of the energy buying and selling prices. Instead of using the averaged value, “Intermediate-Market Rate” relies on a convex linear combination of the minimum retailer selling price and the maximum retailer buying price. “Supply-Demand Ratio” (SDR) is based on the calculation of the ratio between the energy available for self-consumption and the energy demanded by those needing supply. SDR may also include a compensation factor to share the benefit of members that consume when there is an excess of local generation with prosumers that contribute to this local excess. Finally, “Post-Delivery Pool-Based” pricing mechanism is an application of auction theory based on a fair design approach (search for the local electricity market equilibrium prices). The local prices are calculated after delivery considering, for each prosumer, the energy produced and consumed and the corresponding retailer buying and selling prices. The comparison of these mechanisms shows that this last one provides the best price signals relevance and efficiency economic signal with the lowest price volatility and the highest fairness.

However, the performance of these models is heavily dependent on economic considerations related to electricity prices which can be divided into two main categories of tariffs:

Table 2
Comparative analytical properties of ESh paradigms.

Dimension	RB	Centralized Allocation	Distributed Allocation	Market-Based	Hybrid
Allocation logic	Predefined coefficients	Global optimization	Decomposed optimization	Strategic equilibrium	Layered optimization + pricing
Optimization methods	None/Heuristics	LP, QP, MILP, stochastic programming	ADMM, dual-decomposition, proximal methods	Equilibrium models (GNE/VI), auction-clearing algorithms	Bilevel optimization, multi-layer decomposition
Price formation	Exogenous/fixed	Dual shadow prices	Algorithmic dual signals	Endogenous equilibrium prices	Dual + equilibrium interplay
Welfare optimality	No guarantee	Yes (under convexity)	Yes (under convexity)	Only under ideal assumptions	Context-dependent
Incentive compatibility	Weak	Not modelled	Cooperative assumption	Explicitly modelled	Partial
Strategic behaviour	Absent	Absent	Absent	Central	Partial
Fairness integration	Implicit via coefficients	Can be embedded	Can be embedded	Rare/constrained	Multi-layer
Scalability	High	Limited	High	Medium–High	Variable
Regulatory embedding	Strong	Constrained	Constrained	Complex	Highly sensitive

- The external grid prices, also called retail prices, which include the electricity purchase price paid by each member to the retailer—often depending on the subscribed contract (flat, time-of-use, or dynamic)—as well as the selling price at which prosumers export their surplus energy to the grid. These tariffs have been very widely presented in the literature and are not the main focus of our review [134–137].
- The internal community tariffs, which define the price at which electricity is exchanged between members, both for buying and selling within EC. Their design is not only driven by efficiency considerations, but also by distributional objectives. In particular, internal prices can be structured to ensure individual rationality, to respect cost-causation principles, or to promote fairness among participants.

Among the many studies on EC, this section focuses on articles that discuss the different tariffs and, where possible, present the associated benefits.

Based on [138], authors in Ref. [139] generalized six axioms for community pricing framework: individual rationality, profit-neutrality, equity, monotonicity, cost-causation penalty and cost-mitigation reward. Each of them is introduced as a constraint in the optimization problem, ensuring EC welfare. A dynamic pricing mechanism called Dynamic Net Energy Metering (NEM) was also introduced, encouraging EC members to plan their resources in order to achieve optimal overall well-being while respecting the principle of cost causality. This is achieved by introducing a threshold-based, resource-aware structure, announcing EC prices based on the level of aggregate renewable generation. The same authors proposed a more recent work [140], now based on four axioms: uniform payment function, monotonicity and cost-causation, individual rationality and profit neutrality. The operator's pricing objective is to maximize the EC social welfare defined by the total surplus of its members. The method was applied to 23 residential households with flexible loads (18 with PV roofs and shared ESS) and an average gain of 6.32% with ESS and 4.29% without ESS was achieved.

Mustika et al. [90] proposed a two-stage management strategy for the optimal operation and billing in a French EC. In this EC, the price of the electricity sold to the members is slightly higher than the one sold to the grid: 7 c€/kWh and 6.5 c€/kWh respectively. The difference is more pronounced for the buying price: 7.5 c€/kWh for the EC and between 12.97 and 13.31 c€/kWh for the grid, depending on the subscribed power and retailer of each participant. Compared to a case in which the members are not organized in EC, a saving of 11.7% on the global community bill and 11 to 19% of individual savings are observed.

Mena et al. [91] optimized CSC in residential buildings in Spain. They focused on the regulated market price, fixed by the Spanish government; the electricity bill is composed of a fixed (power) part linked to transmission & distribution tolls and a variable (energy) part depending on the period, peak, flat or valley, with corresponding costs: 7.441 c€/kWh, 2.847 c€/kWh and 0.303 c€/kWh. In this EC model, the global renewable energy generated is divided among the participants, allowing the calculation of the electricity bill for each household. The sharing of the net hourly energy generated individually for each participant is established by Annex I of Royal Decree 144/2019. With a whole-year model, a reduction in the cost of energy (COE) value of 15.5 % for SC without surpluses and 51.5 % for SC with surpluses under compensation are obtained.

Llera-Sastresa et al. [29] worked on another Spanish study case to assess the impact of different sharing schemes on CSC. Three types of prosumers were introduced, each with its own electricity tariff depending on the contracted power and the time period (flat: 12.29 c€/kWh, double: 14.517 c€/kWh and 7.398 c€/kWh and triple: 10.146 c€/kWh, 8.794 c€/kWh and 6.460 c€/kWh). Here, prosumers' surpluses are assumed to be repaid on the invoice using a simplified compensation mechanism, at 4.9 c€/kWh. Depending on the energy-sharing criterion selected among the eight tested, a maximum reduction of approximately 40% in the total annual cost can be achieved.

Balegamire and Gautier [141] considered several terms in the EC profit calculation: the revenue from selling the production (to members and grid), the cost of buying electricity from prosumers, investment in storage (if applicable), administrative costs and total membership fee paid by the members. The retail prices from a Belgian supplier are also introduced for the calculation of individual benefit (selling price: 28.1 c€/kWh, buying price: 7.2 c€/kWh) and used as boundaries for the prices within EC. Depending on the objective (equality or efficiency), different pairs of values are proposed. As in Ref. [112], authors also insist that internal prices and sharing rules do not affect the value created, as they are only intended to serve as mechanisms for redistributing value among members.

Gjorgievski et al. [93] proposed a method based on individual contribution, called "virtual net-billing" (VNB) (see Eq. (18)). The VNB method guarantees the conservation of ISC as well as a part of the shared electricity at a given price. This price is bounded by two tariffs: the one for the net-exported electricity (3.6 US'/kWh) and the one for the net-imported electricity (8 US'/kWh); here it is set at the lower bound in order to maximize the EC economic savings.

In [142], the authors focused on external factors influencing the financial benefits of EC, such as regulatory frameworks and fiscal policies; regulated charges in Europe may reduce households' savings from about 62.3% to 18.6% of the monthly electricity bill.

Šironja et al. [143] focused on two Croatian EC models: one with only business entities and the other with a mix of residential and business consumers. The buying price is based on time-of-use (high-tariff during day and low-tariff during night) with different prices for the two types of consumers: 7.9 and 3.9 c€/kWh for households and 35 and 21 c€/kWh for business. The selling price follows the same principle: 6 and 3 c€/kWh for households and 28 and 16 c€/kWh for business. To ensure a fair cost-benefit distribution among participants, the prices of shared energy are set as the mean value between buying and selling prices of the supplier. Croatian regulations also impose a full-time manager for the EC which has not been considered in this study. Depending on the scenario, the two ECs reach, respectively, a mean annual electricity bill reduction of 80.6% and 72.4% with a mean payback period of 2.51 and 8.86 years.

In addition to economic gains, other considerations also arise. Gjorgievski et al. [115] addressed how different electricity tariffs affect both the EC's peak power exchanges with the distribution grid and its energy autonomy. Innovative electricity tariffs are introduced as an effective tool to encourage efficient management of energy sharing. Eight electricity tariffs are considered based on different combinations of energy and supply cost and regulated cost. Half use static energy and supply cost and the other half uses dynamic energy and supply cost. As for the regulated cost for the shared energy, it can be chosen from four methods: i) "business-as-usual" with a reduced cost of 7.13 c€/kWh instead of 15.06 c€/kWh, ii) "pure self-consumption" with the same tariff but no possibility of selling surplus to the main grid, iii) "time--of-use" with a low-cost period (−30% from 0 a.m. to 4 p.m.) and a high-cost period (+30% from 5 p.m. to 11 p.m.) and iv) "energy and capacity payment" which is specific to the case study with a common district heating system and not relevant for this review.

Among the different reviewed studies, the concept of fairness is regularly mentioned but its consideration varies considerably. A brief review on fairness in energy sharing is proposed in Ref. [93] and illustrates the different interpretations in the scientific literature. The same authors use specific indicators such as Jain's index, fairness index or maximum excess to take fairness into account in their analysis. Couraud et al. [113] also used a detailed pricing model to compare the fairness of various energy distribution algorithms. In Ref. [130], the authors targeted the problem of fairness in EC by considering the energy poverty level of the members. As usual, the maximum and minimum prices at which each member is willing to buy and sell electricity are still directly linked to the prices contracted and agreed with the energy retailer. However, a new criterion is added to decrease the unfairness level

between the different groups. Regarding the grid electricity tariffs, these are based on Enovos, a Luxembourgish electricity supplier offering three types of tariffs: flat (18.74 c€/kWh), double (18.99 c€/kWh from 6 a.m. to 10 p.m. and 17.77 c€/kWh for 10 p.m. to 6 p.m.) and dynamic based on EPEX energy market. The supplier also buys the surplus energy at a flat price of 14.17 c€/kWh. As the aim of the proposed method is not to increase the EC revenue, the economic performance is not discussed here.

This brief literature review highlights two key points for pricing and cost allocation in CSC schemes: the critical role of grid electricity costs and retail contract choices, and the diversity of cost calculation methods with a growing emphasis on fairness.

A widely shared principle across the reviewed studies is that internal community tariffs must remain bounded between retail buying and selling prices. Local community prices should be low enough to ensure that consumers benefit compared to purchasing electricity from the grid, while guaranteeing that prosumers receive at least as much as they would by exporting their surplus to the public grid. Regardless of the concept of fairness, this price bounding condition appears as a necessary requirement for a viable energy community. It also makes the costs highly dependent on the case study, particularly influenced by the regulations in force in the country and the date of operation.

The literature also shows that fixed charges linked to subscribed power, regulated transmission and distribution fees can significantly reduce the achievable savings from collective self-consumption. Several studies demonstrate that even highly efficient local energy sharing schemes may deliver limited benefits if regulated charges remain high or poorly adapted to CSC. Consequently, the selection of contracts and the treatment of grid costs for shared energy strongly influence both individual bills and community-level outcomes, sometimes more than the internal pricing rule itself. Although the level of performance is highly sensitive to tariff design and network charges, the literature consistently shows that CSC generates net economic benefits for participants compared to conventional individual supply arrangements.

Finally, a broad range of cost calculation and allocation methods are proposed in the literature, reflecting different objectives such as efficiency or fairness for which a compromise must be found. Increasingly, fairness is treated as a key criterion and some recent works extend fairness considerations beyond purely economic contributions by incorporating social dimensions, such as heterogeneous income levels or energy poverty indicators. These contributions underline that fair cost allocation is essential not only for ethical reasons, but also to ensure long-term acceptability, stability, and cohesion within energy communities.

3. Discussion: practical implications of the unified framework

The unified framework developed in this review has three main practical implications. First, it helps clarify the design choices faced by energy community operators. A community does not only choose whether to share energy or not; it must choose how allocation decisions are made, how prices are formed, how benefits are distributed, and how regulatory constraints are embedded. Rule-based allocation is simple and compatible with many regulatory frameworks, but it does not generally guarantee allocative efficiency. Centralized optimization provides a useful welfare benchmark, but it requires information centralization and raises governance issues. Distributed optimization reduces data centralization but assumes cooperative participation. Market-based mechanisms introduce endogenous price formation and incentive considerations, but they also increase strategic complexity. Hybrid architectures attempt to combine these properties, but they require careful coordination between technical dispatch, economic settlement, and regulatory admissibility.

Second, the framework clarifies the relevance of the review for regulators and policy-oriented readers. Internal tariff rules, grid charges, and admissible settlement mechanisms strongly affect whether energy-sharing schemes create individual savings, preserve fairness, and remain financially viable. In this sense, regulation does not merely constrain implementation after the optimization problem has been solved; it shapes the feasible set of pricing and redistribution mechanisms. A mechanism that is welfare-efficient in theory may be difficult to implement if its internal price signals are incompatible with existing tariff rules, grid-fee structures, or legal definitions of collective self-consumption.

Third, the framework provides a common basis for comparing studies that otherwise use different terminology and performance indicators. Some studies report self-consumption or self-sufficiency gains, others report aggregate welfare, individual bill reductions, fairness indicators, or equilibrium properties. Without a common analytical structure, these results may appear contradictory even when they describe different dimensions of performance. By separating physical feasibility, coordination logic, price formation, and distributive effects, the framework makes these comparisons more transparent.

From a practical standpoint, the framework therefore supports a design-oriented reading of the literature. Small or highly regulated communities may prioritize rule-based or dynamically adjusted coefficients. Larger communities with flexible assets may benefit from centralized or distributed optimization. Communities aiming to promote autonomous transactions between participants may require market-based or hybrid mechanisms, provided that strategic behaviour, fairness, and regulatory compliance are explicitly addressed. The appropriate design is therefore context-dependent and should be selected according to community size, data availability, regulatory environment, flexibility resources, and distributive objectives.

4. Research gaps and future trends

4.1. Research gaps

Despite rapid methodological progress, the literature on ESh mechanisms remains structurally fragmented. A first limitation concerns the lack of conceptual alignment between performance metrics. Technical indicators such as CSC or self-sufficiency are frequently used alongside welfare-based objectives without a systematic articulation of how these criteria interact or conflict. As a result, mechanisms are often evaluated under partially overlapping notions of efficiency, making cross-study comparison difficult and obscuring trade-offs under realistic tariff structures.

More fundamentally, many coordination models rely on behavioural and informational assumptions that simplify away strategic complexity. Centralized and distributed allocation frameworks typically presume cooperative participation and truthful parameter revelation, thereby sidestepping incentive compatibility issues. This creates a theoretical disconnect between optimization-based coordination and the strategic environments in which heterogeneous agents actually operate. The limited integration of mechanism design principles into distributed energy allocation leaves open questions about robustness to manipulation, information asymmetry, and coalition formation.

A similar fragmentation characterizes the treatment of fairness. Equity considerations are introduced either through ex post redistribution, cooperative benchmarks, or additional constraints in optimization problems. However, fairness is rarely embedded simultaneously across dispatch decisions, pricing rules, and regulatory conditions. This layered separation prevents a systematic understanding of how equity objectives reshape equilibrium outcomes, particularly in hybrid architectures where efficiency and distributive justice may conflict.

The gap between theoretical optimality and regulatory feasibility further constrains the external validity of many contributions. Optimization-based studies often abstract from legal constraints governing tariffs, grid fees, and settlement mechanisms. Yet regulatory embedding fundamentally shapes the set of implementable transfers and pricing schemes. The divergence between shadow-price optimality and

legally admissible arrangements remains insufficiently examined, limiting the policy relevance of otherwise rigorous analytical models.

Finally, much of the theoretical equivalence between centralized solutions, distributed algorithms, and market equilibria relies on convexity assumptions that rarely hold in practice. Binary scheduling decisions, minimum operating thresholds, and tariff discontinuities introduce nonconvexities that can undermine equilibrium existence, convergence guarantees, and welfare comparability. At scale, additional challenges emerge in communication overhead, privacy preservation, and data governance, areas where empirical validation across heterogeneous communities remains limited.

Taken together, these gaps indicate that the field has developed sophisticated coordination tools, but lacks a fully integrated analytical framework capable of simultaneously addressing efficiency measurement, strategic behaviour, fairness, regulatory feasibility, and real-world computational constraints.

4.2. Future trends

Future research is likely to shift from proposing isolated coordination paradigms toward systematically integrating efficiency, incentives, fairness, and regulatory feasibility within unified analytical frameworks.

A key direction concerns integrated mechanism design. Rather than treating centralized optimization and market-based coordination as distinct paradigms, future models may jointly design dispatch rules and pricing mechanisms to ensure welfare efficiency, incentive compatibility, and fairness under realistic information structures.

Hybrid multi-layer architectures also represent a promising frontier. Combining centralized optimization, distributed computation, and market-based price signals allows greater flexibility but introduces cross-layer tensions. Formal analysis of dual-equilibrium divergence, regulatory admissibility, and fairness constraints across coordination layers will be essential for real-world implementation.

Another important trend is the development of dynamic and adaptive allocation rules. Moving beyond static rule-based coefficients, future systems may integrate forecasting, learning algorithms, and optimization-based adaptation to respond to evolving demand, generation, and tariff conditions while maintaining regulatory simplicity.

Finally, distributionally robust and equity-aware formulations are likely to gain prominence. Embedding inequality metrics or fairness-oriented constraints directly into equilibrium, distributed, or hybrid coordination models offers a pathway to reconcile social justice objectives with economic efficiency in energy communities.

These research directions jointly reinforce the view that energy sharing should be understood as a continuum of coordination architectures rather than a dichotomy between centralized and market-based paradigms.

5. Conclusion

The unified framework developed in Section 2 allows ESh mechanisms to be interpreted not as heterogeneous institutional arrangements, but as alternative coordination logics applied to a common constrained physical system. This perspective reveals that differences across rule-based, allocation-based, distributed, market-based, and hybrid models arise from the way decision authority, information, and price formation are structured rather than from distinct technical infrastructures.

5.1. Several structural insights emerge

First, allocative efficiency is not an intrinsic property of a mechanism but depends on the coordination architecture and underlying assumptions. Centralized allocation provides a normative welfare benchmark under convexity and full information. Distributed optimization preserves this benchmark under cooperative behaviour and well-behaved cost structures. Market-based mechanisms can replicate welfare optimality only under ideal competitive assumptions. Rule-based schemes, by contrast, restrict the feasible allocation space ex ante and generally sacrifice allocative efficiency for administrative simplicity and regulatory compatibility.

Second, price formation plays a pivotal differentiating role. Dual variables in centralized and distributed allocation act as coordination signals but do not emerge from strategic interaction. In contrast, market-based systems internalize endogenous price formation and incentive compatibility at the cost of strategic complexity and potential welfare loss. Hybrid models attempt to reconcile these dimensions but introduce cross-layer tensions between shadow prices, equilibrium prices, and regulatory constraints.

Third, efficiency, fairness, scalability, and regulatory embedding are not orthogonal dimensions. Improving one dimension often weakens another. For example, fairness-constrained optimization may reduce aggregate surplus; endogenous pricing enhances incentive compatibility but increases strategic behaviour and regulatory oversight requirements; distributed decomposition improves scalability while assuming cooperative participation.

Taken together, these results suggest that energy sharing paradigms should be evaluated as institutional and mathematical design choices along a coordination spectrum rather than as competing technologies.

CRediT authorship contribution statement

Adrien Chatel: Conceptualization, Methodology, Writing – Original draft. Gilles Notton: Conceptualization, Supervision, Writing – Original draft. Ghjuvan Antone Faggianelli: Conceptualization, Supervision, Writing – Original draft. Cyril Voyant: Supervision, Writing – Original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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